



Assessing labor market frictions in a small open economy[☆]



Jeffrey Sheen^a, Ben Zhe Wang^{a,*}

Department of Economics, Macquarie University, NSW, 2109, Australia

ARTICLE INFO

Article history:

Received 17 November 2015

Revised 24 February 2016

Accepted 25 February 2016

Available online 19 April 2016

JEL classification:

C11

C13

E32

F41

J64.

Keywords:

DSGE

Hiring costs

Wage rigidities

Bayesian estimation

Small open economy

ABSTRACT

We estimate small open economy models with involuntary unemployment using Australian data from 1993 to 2013, assessing whether nominal or real wage rigidity and hiring costs matter. Nominal wage rigidity with hiring costs (accounting for 1.58% of GDP) are strongly preferred. Using the preferred model, technology shocks explain most short-run unemployment and long run real wage variance. External shocks have important lasting effects on inflation and consumption, and on long-run investment and interest rates, but little on the labor market. Out-of-sample conditional forecasts perform well but cannot predict the confidence effects of the crisis in 2008.

© 2016 Elsevier Inc. All rights reserved.

1. Introduction

We estimate medium-scale small open economy models with labor market frictions using Australian data from 1993Q2 to 2013Q2. The small open economy features of the models are based on [Adolfson et al. \(2007\)](#), who analyze the Euro economies by incorporating various nominal and real rigidities including nominal price and wage rigidities, costly investment adjustment, habit formation and external risk premia. Further, we follow [Blanchard and Gali \(2010\)](#) by characterizing the labor market as searching and matching between firms and employees. In each period, there is a fixed chance of a worker separating from existing employment, and firms have to pay a hiring cost to employ a new worker. These hiring costs depend on the current labor market conditions, which in turn depend on the lagged and current unemployment. Forward-looking firms internalize this cost when making price decisions, and thus inflation depends on lagged, current and future expected unemployment rate. These various labor market frictions permit involuntary unemployment in our small open economy model and create a monetary policy trade-off between output(or unemployment) and inflation, thus

[☆] We would like to thank the editors of the journal, James Morley, Jarkko Jääskelä, Olivier Blanchard, Doug Laxton and Michael Kumhof for useful suggestions on previous versions of this paper. We thank participants at the Reserve Bank of New Zealand, the 2012 Methods of International Finance Network Conference, the 2012 Workshop in Macroeconomic Dynamics and the 2013 Dynare conference.

* Corresponding author. Tel: +61 2 9850 8500.

E-mail addresses: jeffrey.sheen@mq.edu.au (J. Sheen), ben.wang@mq.edu.au (B.Z. Wang).

avoiding the so-called ‘divine coincidence’.¹ Further, equilibrium real wages are wedged by the hiring costs. Shimer (2005) argues that real wages determined by Nash bargaining cannot generate the observed fluctuations of unemployment. To deal with this, we follow Hall (2005) in allowing a degree of real wage rigidity. On the other hand, Riggi (2010) and Riggi and Tancioni (2010) argue that real wage rigidity is unlikely to account for the productivity–employment puzzle—facing a positive technology shock, a model with real wage rigidity tends to exhibit a rise in employment, which contradicts most empirical findings as discussed in Gali (1999), Francis and Ramey (2005) and Basu et al. (2006).

Given this contest of ideas, our contributions to the literature are three-fold. Firstly, we test whether our data set supports the existence of hiring costs, and which type of wage rigidity is preferred, given a unified structural macroeconomic model. Secondly, we evaluate how these labor market assumptions impact on parameter estimates and thus the key features of the estimated New Keynesian macroeconomic model. Lastly, we evaluate the out-of-sample performance of our preferred model for the Australian economy through the recent global crisis to see whether our mainstream model remains valid.

Using Bayesian methods, our results indicate that the model with hiring costs and nominal wage rigidity fits the data best. We estimate that hiring costs accounted for a mean of 1.58% of GDP, which is broadly consistent with the Australian Bureau of Statistics estimates that the total industry value-added by employment placement organisations contributed 1.3% to Australian gross domestic product (GDP) in the year 2001–02.² Using Bayes’ factor as the guide, our results show that nominal wage rigidity is preferred to real wage rigidity regardless of the existence of hiring costs.

Decomposing variances using our preferred model reveals that technology shocks, which drive labor demand and incur hiring costs, dominate unemployment in the short run. Labor preference shocks have minimal impact on unemployment variance in the short run, but play a moderate role for real wage variance. In the long run, we find both technology and labor preference shocks are important in explaining unemployment and real wage variances. In particular, technology shocks explain nearly half of the unemployment variance and more than half of the real wage variance. Consistent with our description of the models as New Keynesian, we find that output variances are mostly explained by aggregate demand shocks in the short run, and by aggregate supply shocks in the long run. We find external shocks to this small open economy have important lasting effects on inflation and consumption variance, and significant effects on investment and interest rates only in the long run. However, there are minimal impacts of external shocks on the labour market in the short and long run.

To evaluate the performance of our preferred model, we obtain out-of-sample conditional forecasts from the end of 2007 through to 2013. Treating external variables as exogenous, the model performs remarkably well, with modest root mean square errors for domestic macroeconomic variables. Our results indicate that our mainstream New Keynesian macroeconomic model remains valid in the case of Australia. A new macroeconomic paradigm is not needed in this case, which is not surprising given that Australia was not heavily affected by the crisis.

In regard to our model, there are several related papers. Early attempts at estimating DSGE models for Australia include Buncic and Melecky (2008) and Nimark (2009), who construct and estimate a version of a small open economy New Keynesian model. Jääskelä and Nimark (2011) estimate a version of Adolfson et al. (2007) in a data-rich environment by incorporating the commodity market. However, these authors did not directly consider involuntary unemployment. Faccini et al. (2013) estimate a model with matching frictions and nominal wage rigidity for the UK economy, and they find nominal wage rigidity improves the model fit but are irrelevant for inflation dynamics at the estimated equilibrium. Using New Zealand data, Albertini et al. (2012) estimate the small open economy model of Gali and Monacelli (2005) with search and matching frictions, and find the majority of variation in the New Zealand labor market is solely explained by disturbances pertaining to the labor market. Christiano et al. (2011) extend the small open economy model of Adolfson et al. (2007) to include financial and labor market frictions (focusing on the former, and assuming nominal wage rigidity). Their model is estimated using Swedish data, and they find the hiring cost accounts for about 0.39% of GDP and that labor preference shock plays an important role in driving wages (but not GDP and unemployment). For the U.S., Gertler et al. (2008) estimate a model with indivisible, costly-to-adjust labor and staggered nominal wage bargaining, and find that nominal wage rigidity fits the data well compared to the flexible wage version. Gali et al. (2011) extend Smets and Wouters (2007) with unemployment, and estimate the resulting model with unemployment data.

This paper is organized as follows. Section 2 outlines the model. Section 3 discusses the estimation procedures, Section 3.2 presents the comparison of alternative models, Section 4 shows the detailed posterior estimation, impulse responses and variance decomposition for our preferred model, and Section 5 concludes.

2. Model

Our model extends Adolfson et al. (2007), with labor market frictions as discussed in Blanchard and Gali (2010).³ The economy consists of four types of representative firms: domestic goods-producing firms, consumption-importing firms,

¹ The concept, ‘divine coincidence’, was introduced by Blanchard and Gali (2007) who explained that it occurs in simple New Keynesian models where stabilizing inflation automatically stabilizes output, thus justifying simple inflation targeting.

² From ABS Catalogue number 8558, Employment Services, Australia 2001–02.

³ We depart from Adolfson et al. (2007), in the following ways: 1. we assume firms have to finance all of their wage bill by borrowing at the nominal cash rate; and 2. the government is assumed to distribute all tax revenue to households as a lump sum payment. We recognize the important role played by fiscal policy, especially during recessions and financial distress, and we leave that for future work.

investment-importing firms and exporters. Households derive utility from both domestic and imported consumption, subject to habit formation, and also enjoy leisure and real balances.

2.1. Domestic goods-producing firms

2.1.1. Final domestic goods

A final producer can costlessly combine intermediate goods, transforming them into a homogeneous good that can be used for consumption and investment, in both domestic and overseas economies. Domestic-produced goods compete with imported goods bought by the domestic-importing firms, and the domestic-exporting firms export a fraction of the final product abroad. Final goods are aggregated by:

$$Y_t = \left[\int_0^1 Y_{i,t}^{\frac{1}{\lambda_t^d}} di \right]^{\lambda_t^d}, \quad 1 \leq \lambda_t^d < \infty$$

where $Y_{i,t}$ denotes purchased intermediate product with $i \in [0, 1]$. λ_t^d is the markup for the domestic goods market, which is assumed to follow a persistent stochastic process:

$$\lambda_t^d = (1 - \rho_{\lambda^d}) \bar{\lambda}^d + \rho_{\lambda^d} \lambda_{t-1}^d + \epsilon_t^{\lambda^d}$$

where $\bar{\lambda}^d$ is the steady-state level of markup, ρ_{λ^d} measures persistence and $\epsilon_t^{\lambda^d}$ is an i.i.d. shock. Profit maximization by the final-producing firm yields the following demand curve for the intermediate product i :

$$Y_{i,t} = \left[\frac{P_t^d}{P_{i,t}^d} \right]^{\frac{\lambda_t^d}{\lambda_t^d - 1}} Y_t \quad (2.1)$$

where $P_{i,t}^d$ is the output price of intermediate firm i , and P_t^d is the price of the final good compiled as:

$$P_t^d = \left[\int_0^1 (P_{i,t}^d)^{\frac{1}{1-\lambda_t^d}} di \right]^{1-\lambda_t^d}$$

2.1.2. Intermediate domestic goods

Intermediate firms are subject to monopolistic competition, with each using the following production technology:

$$Y_{i,t} = z_t^{1-\alpha} \epsilon_t K_{i,t}^\alpha N_{i,t}^{1-\alpha} - z_t \phi \quad (2.2)$$

ϕ is a 'fixed' cost scaled by the permanent technology shock z_t to ensure it grows at the economy-wide balanced growth steady state rate, and does not vanish with economic growth. $K_{i,t}$ and $N_{i,t}$ are the capital and labor services used in the production process, and z_t and ϵ_t are permanent and temporary technology shocks with log-deviation processes:

$$\log \left(\frac{z_t}{z_{t-1}} \right) \equiv \hat{\mu}_t^z = \rho_{\mu^z} \mu_{t-1}^z + \epsilon_t^z$$

$$\hat{\epsilon}_t = \rho_\epsilon \hat{\epsilon}_{t-1} + \epsilon_t^\epsilon$$

$\mu_t^z = \frac{z_t}{z_{t-1}}$ is the source of the growth rate of real variables, ϵ_t^z and ϵ_t^ϵ are i.i.d. innovations and ρ_{μ^z} and ρ_ϵ are persistence parameters.

Hiring costs

Following Blanchard and Gali (2010), a fraction δ of the labor force is separated from existing employment in each period. Intermediate firms incur a real cost G_t if they hire a new worker:⁴

$$G_t = z_t \epsilon_t B x_t^\theta \zeta_t^x \quad (2.3)$$

where B is a positive constant that measures the scale of the hiring cost. ζ_t^x is an AR(1) hiring cost shock that follows in log deviation form:

$$\hat{\zeta}_t^x = \rho_x \hat{\zeta}_{t-1}^x + \epsilon_t^x$$

The hiring cost depends on current labor market tightness x_t , defined as the ratio of the number of hires H_t to the number of unemployed before hiring proceeds at t :

$$x_t \equiv \frac{H_t}{(U_{t-1} + \delta N_{t-1})}$$

⁴ The hiring cost is indexed to the permanent technology shock to ensure it grows at the same rate as other real variables in the balanced growth steady state. Similar to the fixed cost, we assume constant return to scale for the hiring cost. Non-constant returns to scale are possible, but should not matter much for the evolution of major macroeconomic variables. This assumption may be better tested in a partial equilibrium model and thus we leave it for future research.

where $H_t \equiv \int_0^1 H_{i,t} di$ is aggregate hiring, $N_t \equiv \int_0^1 N_{i,t} di$ denotes aggregate employment, and $U_{t-1} = 1 - N_{t-1}$ is the number of unemployed *after* hiring ends in $t-1$, assuming the labor force is normalized at 1. Note x_t can be also interpreted as the job-finding rate from the perspective of the unemployed. The evolution of hiring is given by:

$$H_t = N_t - (1 - \delta)N_{t-1}$$

where the number of hires is the difference between current and lagged employment, after accounting for separation. Note G_t is indexed to technology, so it grows with other real variables.⁵ It is convenient to work with the stationary hiring cost:

$$g_t = \frac{G_t}{z_t} = \epsilon_t B x_t^\gamma \zeta_t^x$$

Optimal factor usage

Assume the intermediate firm i has to borrow in advance to fund its wage bill.⁶ Therefore its nominal wage bill is $R_{t-1}W_tN_{i,t}$ at time t , where W_t is the nominal wage and R_{t-1} is the gross nominal interest rate. To optimally allocate resources across factors, firm i minimizes its expected present discounted cost subject to the production technology (2.2):

$$\min_{K_{i,t}, N_{i,t}} \sum_{s=0}^{\infty} \beta^s (R_{t-1}W_tN_{i,t} + P_t^d G_t H_{i,t} + R_t^k K_{i,t} + \lambda_t P_{i,t}^d [Y_{i,t} - z_t^{1-\alpha} \epsilon_t K_{i,t}^\alpha N_{i,t}^{1-\alpha} + z_t \phi])$$

where R_t^k is the gross nominal rental rate on capital service and λ_t is the Lagrange multiplier. The permanent technology shock implies that all real variables in the model have a stochastic trend. Let $k_{t+1} \equiv \frac{K_{t+1}}{z_t}$ represent the de-trended (pre-determined) capital service stock. In addition, since the price level is not stationary, nominal variables will share a nominal stochastic trend. To stationarize the model, let $w_t \equiv \frac{W_t}{P_t^d z_t}$ denote the de-trended real producer wage rate, and $r_t^k \equiv \frac{R_t^k}{P_t^d}$ as the real rental rate of capital earned by households, and define the gross domestic inflation rate as $\pi_t^d \equiv \frac{P_t^d}{P_{t-1}^d}$. The real marginal cost for an intermediate producer, mc_t^d , can be defined as $mc_t^d = \lambda_t \frac{P_{i,t}^d}{P_t^d}$, which from the first order conditions becomes:

$$mc_t^d = \left(\frac{1}{1-\alpha} \right)^{1-\alpha} \left(\frac{1}{\alpha} \right)^\alpha \left(\frac{r_t^k}{\epsilon_t} \right)^\alpha (R_{t-1}w_t + g_t - \beta(1-\delta)\mu_{t+1}^z \pi_{t+1}^d g_{t+1})^{1-\alpha} \quad (2.4)$$

The real marginal cost depends on the rental rate of capital, wage rate, and both current and future expected hiring costs. This cost will appear subsequently in the new Keynesian Phillips curve (NKPC), which links labor market conditions with inflation.

Price setting

Following Calvo (1983), each intermediate firm faces a random probability ξ_d of not receiving a ‘price signal’ to re-optimize its price at $P_{t,new}^d$ in period t , in which case it simply indexes its price to the intermediate firm inflation rate π_{t-1}^d and the perceived central bank inflation target π_t^T , which is defined in Section 2.4:

$$P_{t,idx}^d = (\pi_{t-1}^d)^{\kappa_d} (\pi_t^T)^{1-\kappa_d} P_{t-1}^d$$

where κ_d is the degree of the indexation to past inflation by intermediate firms. Otherwise, with probability $(1 - \xi_d)$, the intermediate firm i receives a ‘price signal’ to re-optimize, and so it sets its price at the level that maximizes its expected discounted profit:

$$\max_{P_{t,new}^d} E_t \sum_{s=0}^{\infty} (\beta \xi_d)^s v_{t+s} [(\pi_t^d \pi_{t+1}^d \dots \pi_{t+s-1}^d)^{\kappa_d} (\pi_{t+1}^T \pi_{t+2}^T \dots \pi_{t+s}^T)^{1-\kappa_d} P_{t,new}^d Y_{i,t+s} - MC_{i,t+s}^d (Y_{i,t+s} + z_{t+s} \phi)] \quad (2.5)$$

where $MC_{t+s}^d = mc_{t+s}^d P_{t+s}^d$ represents the nominal marginal cost and $(\beta \xi_d)^s v_{t+s}$ is the stochastic discount factor.⁷ Substituting the demand for intermediate good i , (2.1) into (2.5), differentiating with respect to $P_{t,new}^d$, setting that to 0, and then linearizing around the steady state gives the optimal price-setting rule when an intermediate firm re-optimizes its price.

The aggregate price index is a weighted-average of the indexed and re-optimized intermediate prices:

$$P_t^d = \left[\xi_d \left((\pi_{t-1}^d)^{\kappa_d} (\pi_t^T)^{1-\kappa_d} P_{t-1}^d \right)^{\frac{1}{1-\lambda_t^d}} + (1 - \xi_d) \left(P_{t,new}^d \right)^{\frac{1}{1-\lambda_t^d}} \right]^{1-\lambda_t^d}$$

⁵ This assumption that a permanent technology shock immediately raises the hiring cost, one-for-one, implies that additional hiring is delayed until the capital stock has adjusted to the new permanently higher level. If we allowed a delay in the impact effect on hiring costs, the hiring impact would be higher after such a shock. However it is hard to justify such a delay. The presence of the temporary technology shock in the hiring cost aligns the effects of this shock on production and hiring costs, which in turn keeps the ratio of the hiring to production costs constant.

⁶ Borrowing in advance to fund the wage bill simply delays the effect of monetary policy changes on marginal cost and thus unemployment.

⁷ Since households own firms, profits are valued at the household’s discounted marginal utility of income, $\beta^s v_{t+s}$.

Linearizing this and combining it with the re-optimized price gives the NKPC for the domestic goods market:

$$\hat{\pi}_t^d - \hat{\pi}_t^T = \frac{\beta}{1 + \beta\kappa_d} E_t[\hat{\pi}_{t+1}^d - \rho_\pi \hat{\pi}_t^T] + \frac{\kappa_d}{1 + \beta\kappa_d} [\hat{\pi}_{t-1}^d - \hat{\pi}_t^T] - \frac{\kappa_d \beta (1 - \rho_\pi)}{1 + \beta\kappa_d} \hat{\pi}_t^T + \frac{(1 - \xi_d)(1 - \beta\xi_d)}{\xi_d(1 + \beta\kappa_d)} [\hat{\lambda}_t^d + \widehat{mc}_t^d] \quad (2.6)$$

2.1.3. Domestic-importing firms

Importing firms buy homogeneous goods at the given world price, and transform them into differentiated consumption C^m and investment I^m goods. Incomplete price pass-through arises because importing firms also follow Calvo (1983) pricing. Both aggregate imported consumption and investment goods are a composite of a continuum of i differentiated imported goods through a CES function. Let $IM_t \in \{C_t^m, I_t^m\}$ denotes the aggregate quantity for consumption and investment imports, with IM_{it} the imports of firm i with respective index $a \in \{cm, im\}$, where:

$$IM_t = \left[\int_0^1 (IM_{it})^{\frac{1}{\lambda_t^a}} di \right]^{\lambda_t^a}, \quad 1 \leq \lambda_t^a < \infty$$

Profit maximization ensures the demand for imported consumption good i is:

$$IM_{it} = \left(\frac{P_t^a}{P_{i,t}^a} \right)^{\frac{\lambda_t^a}{\lambda_t^a - 1}} IM_t$$

where P_t^a denotes the aggregate price of the imported goods and $P_{i,t}^a$ is the price of the imported goods set by the firm i . The time-varying markup λ_t^a follows:

$$\lambda_t^a = (1 - \rho_{\lambda^a}) \bar{\lambda}^a + \rho_{\lambda^a} \lambda_{t-1}^a + \epsilon_t^{\lambda^a}$$

In each period, only $1 - \xi_a$ portion of the importing firms can re-optimize their price. For the rest ξ_a , they simply index their price to the respective import inflation and the inflation target: $P_t^a = (\pi_{t-1}^a)^{\kappa_a} (\hat{\pi}_t^T)^{1-\kappa_a} P_{t-1}^a$. Denote S_t^n as the nominal exchange rate (domestic currency per unit of foreign currency) and P^* as the foreign aggregate price level. Since importing firms can differentiate and repack imported goods costlessly, the marginal cost is the relative price between the cost and selling price: $mc_t^a = \frac{P_t^* S_t^n}{P_t^a}$. Similar to the domestic intermediate firms, profit maximizing leads to a New Keynesian Phillips curve for import prices:

$$\begin{aligned} \hat{\pi}_t^a - \hat{\pi}_t^T &= \frac{\beta}{1 + \beta\kappa_a} E_t[\hat{\pi}_{t+1}^a - \rho_\pi \hat{\pi}_t^T] + \frac{\kappa_a}{1 + \beta\kappa_a} [\hat{\pi}_{t-1}^a - \hat{\pi}_t^T] \\ &\quad - \frac{\kappa_a \beta (1 - \rho_\pi)}{1 + \beta\kappa_a} \hat{\pi}_t^T + \frac{(1 - \xi_a)(1 - \beta\xi_a)}{\xi_a(1 + \beta\kappa_a)} [\hat{\lambda}_t^a + \widehat{mc}_t^a] \end{aligned}$$

2.1.4. Domestic-exporting firms

Export firms buy domestic final goods at price P_t^d , differentiate and sell them abroad. Firm i faces the following demand for its product, $X_{i,t}$:

$$X_{i,t} = \left(\frac{P_t^x}{P_{i,t}^x} \right)^{\frac{\lambda_t^x}{\lambda_t^x - 1}} X_t$$

where P_t^x is the aggregate foreign currency price for exports, $P_{i,t}^x$ is the price set by firm i and X_t is the total export volume. The stochastic time-varying markup for export firms is:

$$\lambda_t^x = (1 - \rho_{\lambda^x}) \bar{\lambda}^x + \rho_{\lambda^x} \lambda_{t-1}^x + \epsilon_t^{\lambda^x}$$

Given the domestic economy is small, the total demand for domestic exports is:

$$X_t = \left[\frac{P_t^x}{P_t^*} \right]^{-\eta_f} Y_t^* \quad (2.7)$$

where Y_t^* is foreign output, η_f is the elasticity of substitution in the foreign economy, and P_t^* is the aggregate foreign price level. As for importing firms, with only $1 - \xi_x$ probability can an export firm re-optimize its price to $P_{t,new}^x$ in each period, and for the rest, firms index their price to past export inflation and the inflation target: $P_t^x = (\pi_{t-1}^x)^{\kappa_x} (\hat{\pi}_t^T)^{1-\kappa_x} P_{t-1}^x$. Since no labor and capital services are employed, the marginal cost for the exporting firms is $mc_t^x = \frac{P_t^d}{S_t^n P_t^x}$. Profit maximization leads to a New Keynesian Phillips curve for export prices:

$$\begin{aligned} \hat{\pi}_t^x - \hat{\pi}_t^T &= \frac{\beta}{1 + \beta\kappa_x} E_t[\hat{\pi}_{t+1}^x - \rho_\pi \hat{\pi}_t^T] + \frac{\kappa_x}{1 + \beta\kappa_x} [\hat{\pi}_{t-1}^x - \hat{\pi}_t^T] \\ &\quad - \frac{\kappa_x \beta (1 - \rho_\pi)}{1 + \beta\kappa_x} \hat{\pi}_t^T + \frac{(1 - \xi_x)(1 - \beta\xi_x)}{\xi_x(1 + \beta\kappa_x)} [\hat{\lambda}_t^x + \widehat{mc}_t^x] \end{aligned}$$

2.2. Households

A representative household j with a continuum of members indexed by $\iota \in [0, 1]$ maximizes the following utility function:

$$E_0^j \sum_{t=0}^{\infty} \beta^t \left[\zeta_t^c \ln(C_{j,t} - bC_{j,t-1}) - \zeta_t^N A_L \frac{N_{j,t}^{1+\sigma_L}}{1+\sigma_L} + A_q \frac{\left(\frac{Q_{j,t}}{z_t p_t^c}\right)^{1-\sigma_q}}{1-\sigma_q} \right] \quad (2.8)$$

Households get utility from habitual consumption $C_{j,t}$, with the degree of habit persistence measured by b , disutility from work,⁸ and utility from real cash balances scaled by the permanent technology shock $\frac{Q_{j,t}}{z_t p_t^c}$. ζ_t^l for $l \in \{c, N\}$ are preference shocks for consumption and labor supply, with a steady-state value $E[\zeta_t^l] = 1$.

Following Gertler et al. (2008) and Gali et al. (2011), we model labor so that it varies on its extensive rather than intensive margin (as assumed in Adolfson et al. (2007) and Jääskelä and Nimark (2011)). This turns out to be broadly consistent with Australian experience during the period we study, since employment and hours worked are highly correlated.⁹

The budget constraint for the representative household is:

$$\begin{aligned} R_{t-1}(M_t - Q_t) + Q_t + \Pi_t + W_t N_t + R_t^k u_t \tilde{K}_t + R_{t-1}^* \Phi\left(\frac{A_{t-1}}{z_{t-1}}, \tilde{\phi}_{t-1}\right) S_t^n B_t^* \\ = M_{t+1} + S_{t+1}^n B_{t+1}^* + P_t^c C_t + P_t^I I_t + P_t^d (\tilde{a}(u_t) \tilde{K}_t + P_t^{k'} \Delta_t) \end{aligned} \quad (2.9)$$

where M_t is total nominal domestic assets (either in the form of cash Q_t or domestic interest earning assets $(M_t - Q_t)$). B_t^* is foreign bonds held by the household at time t earning R_t^* . Π_t is profits from domestic firms.¹⁰

The household has three options in varying capital services: 1. purchase new investment goods at price P_t^I , with a cost of installation; or 2. pay the capital utilization cost $\tilde{a}(u_{j,t})$ to change the utilization rate of existing capital stock, where the capital utilization cost is assumed to satisfy $\tilde{a}(1) = 0$, $\tilde{a}'(1) = \bar{r}^k$ and $\tilde{a}''(1) \geq 0$ ¹¹; or 3. go to the market to purchase existing installed capital stock at cost $P_t^{k'} \Delta_t$.

The function $\Phi(\cdot)$ represents the risk premium on holding foreign bonds, which depends on the real aggregate net foreign asset position of the domestic economy $A_t \equiv \frac{S_t^n B_{t+1}^*}{P_t^d}$:

$$\Phi\left(\frac{A_t}{z_t}, \tilde{\phi}_t\right) = \Phi(a_t, \tilde{\phi}_t) = \exp(-\tilde{\phi}_a(a_t - \bar{a}) + \tilde{\phi}_t)$$

where $\tilde{\phi}_t$ is a time-varying shock to the risk premium.¹² The function $\Phi(\cdot)$ is assumed to be strictly decreasing in a_t , so that domestic households are charged a premium on the foreign interest rate if the home country is a net borrower ($B_{t+1}^* < 0$). This risk premium ensures a well-defined steady state in the model (for details see Schmitt-Grohe and Uribe (2003)).

2.2.1. Consumption and investment

Households derive utility from an aggregate of domestic and imported consumption goods:

$$C_t = \left[(1 - \omega_c)^{\frac{1}{\eta_c}} (C_t^d)^{\frac{\eta_c - 1}{\eta_c}} + \omega_c^{\frac{1}{\eta_c}} (C_t^m)^{\frac{\eta_c - 1}{\eta_c}} \right]^{\frac{\eta_c}{\eta_c - 1}}$$

where ω_c is the share of imports in aggregate consumption and η_c is the elasticity of substitution between domestic and foreign consumption goods. Optimal consumption is:

$$C_t^d = (1 - \omega_c) \left[\frac{P_t^d}{P_t^c} \right]^{-\eta_c} C_t \quad (2.10)$$

$$C_t^m = \omega_c \left[\frac{P_t^{m,c}}{P_t^c} \right]^{-\eta_c} C_t \quad (2.11)$$

⁸ A household member ι can either be employed or unemployed. If employed, she incurs a disutility of $A_L \zeta_t^N \iota^{\sigma_L}$ and 0 otherwise, where A_L is a positive constant. Since agents are ex ante homogeneous, integrating over the members who are employed gives $\int_0^{N_{j,t}} A_L \zeta_t^N \iota^{\sigma_L} d\iota = A_L \zeta_t^N \frac{N_{j,t}^{1+\sigma_L}}{1+\sigma_L}$. Note that each member of the household cares about household utility, and so, although each person is fully insured and consumes the same whether employed or not, there will be no free riding.

⁹ Consistent with this assumption, we measure nominal wages in our estimation as the average full-time weekly earnings per adult. Therefore when we consider nominal wage rigidity, our measure does not introduce undue flexibility from variations in hours across the business cycle.

¹⁰ ϕ is set to ensure the steady-state value of this profit equals to 0.

¹¹ Recall $u_t = \frac{K_t}{\tilde{K}_t}$ is the utilization rate with a steady-state value of 1.

¹² In the case of no borrowing at the steady state, $\Phi(0, 0) = 1$.

where P_t^c is the aggregate price for consumption goods (CPI):

$$P_t^c = \left[(1 - \omega_c)(P_t^d)^{1-\eta_c} + \omega_c(P_t^{m,c})^{1-\eta_c} \right]^{\frac{1}{1-\eta_c}}$$

Similarly, households make investment decisions, with aggregate investment being:

$$I_t = \left[(1 - \omega_i)(I_t^d)^{\frac{\eta_i-1}{\eta_i}} + \omega_i(I_t^m)^{\frac{\eta_i-1}{\eta_i}} \right]^{\frac{\eta_i}{\eta_i-1}}$$

The aggregate investment price level P_t^i can deviate from the aggregate consumption price level P_t^c :

$$P_t^i = \left[(1 - \omega_i)(P_t^d)^{1-\eta_i} + \omega_i(P_t^{m,i})^{1-\eta_i} \right]^{\frac{1}{1-\eta_i}}$$

The allocation of demand between domestic and imported investment is:

$$I_t^d = (1 - \omega_i) \left[\frac{P_t^d}{P_t^i} \right]^{-\eta_i} I_t \quad (2.12)$$

$$I_t^m = \omega_i \left[\frac{P_t^{m,i}}{P_t^i} \right]^{-\eta_i} I_t \quad (2.13)$$

Aggregate imports is the sum of consumption and investment imports:

$$M_t^m = C_t^m + I_t^m$$

2.2.2. Capital accumulation

The accumulation of physical capital stock by households $\tilde{K}_{t+1}$ is governed by depreciation δ_k and new investments with an installation cost.

$$\tilde{K}_{t+1} = (1 - \delta_k)\tilde{K}_t + \Gamma_t \left[1 - \tilde{S}\left(\frac{I_t}{I_{t-1}}\right) \right] I_t + \Delta_t$$

where Γ_t is a stationary investment-specific technology shock with $E[\Gamma_t] = 1$. Δ_t reflects households with access to a market where they can purchase new, installed capital $\tilde{K}_{t+1}$ from other households. Since households are representative, $\Delta_t = 0$ in equilibrium. The existence of this market facilitates calculation of the price of second-hand physical capital, $P_t^{k'}$. $1 - \tilde{S}\left(\frac{I_t}{I_{t-1}}\right)$ describes the installation technology, with $\tilde{S}(\bar{\mu}^z) = \tilde{S}'(\bar{\mu}^z) = 0$ and $\tilde{S}''(\mu_t^z) = \tilde{S}'' > 0$. Since the capital stock is pre-determined and has a trend component, we stationarize it by dividing by z_t on both sides of the above equation to give:

$$\tilde{k}_{t+1} = (1 - \delta_k) \frac{\tilde{k}_t}{\mu_t^z} + \Gamma_t f(i_t, i_{t-1}, \mu_t^z) + \frac{\Delta_t}{z_t} \quad (2.14)$$

The household j maximizes the utility function (2.8), subject to the budget constraint (2.9) and the capital accumulation identity (2.14) with respect to c_t , m_{t+1} , Δ_t , $\tilde{k}_{t+1}$, i_t , u_t , q_t and b_{t+1}^* .

The optimal decisions on domestic asset and foreign bond holding yield a risk-adjusted uncovered interest rate parity condition:

$$\hat{R}_t - \hat{R}_t^* = E_t[\hat{S}_{t+1}^n - \hat{S}_t^n] - \tilde{\phi}_a \hat{a}_t + \hat{\phi}_t$$

where $\hat{\phi}_t$ is the time-varying shock to the risk premium, which is assumed to follow a stationary AR(1) process.

2.3. Wage determination under alternative rigidity assumptions

We consider real and nominal wage rigidities as alternatives. We construct real wage rigidity in the spirit of Hall (2003; 2005), where the real wage is a weighted average of the lagged real wage and the equilibrium Nash-bargaining wage. We construct nominal wage rigidity with labor assumed differentiated and organized possibly through unions, and following the Calvo (1983)-type of wage-setting.

2.3.1. Real wage rigidity

For this alternative, we begin by assuming that in the absence of any rigidity, the equilibrium real wage would be determined by Nash bargaining. In each period, both firms and workers calculate the surplus from an established employment relationship. Because firms can replace any worker immediately after paying a hiring cost, the stationarized surplus for a firm is the hiring cost:

$$S_t^F = g_t = B \epsilon_t x_t^\theta \zeta_t^x$$

The worker's surplus from the existing match depends on the wage rate, the opportunity cost measured by the marginal rate of substitution and the future discounted prospect of the existing relationship:¹³

$$V_t^N = w_t - \frac{\zeta_t^N A_L N_t^{\sigma_L}}{\psi_t^z} + \beta E_t \left[\frac{\psi_{t+1}^z}{\psi_t^z \mu_{t+1}^z \pi_{t+1}^d} \left\{ (1 - \delta(1 - x_{t+1})) V_{t+1}^N + \delta(1 - x_{t+1}) V_{t+1}^U \right\} \right] \quad (2.15)$$

which is measured in terms of domestic goods. The term $\beta E_t \left[\frac{\psi_{t+1}^z}{\psi_t^z \mu_{t+1}^z \pi_{t+1}^d} \right]$ is the stochastic discount factor, while x_{t+1} can be interpreted as the job-finding rate in period $t + 1$. The value (again in terms of domestic goods) of a worker who is unemployed is given by:

$$V_t^U = \beta E_t \left[\frac{\psi_{t+1}^z}{\psi_t^z \mu_{t+1}^z \pi_{t+1}^d} \left\{ x_{t+1} V_{t+1}^N + (1 - x_{t+1}) V_{t+1}^U \right\} \right] \quad (2.16)$$

Therefore the net surplus measured in terms of consumption goods to the worker from an existing relationship is the difference between (2.15) and (2.16) adjusted for the relative price of domestic to consumption goods:

$$S_t^H \equiv (V_t^N - V_t^U) \frac{P_t^d}{P_t^c} = w_t - \frac{\zeta_t^N A_L N_t^{\sigma_L}}{\psi_t^z} + \beta(1 - \delta) E_t \left[\frac{\psi_{t+1}^z}{\psi_t^z \mu_{t+1}^z \pi_{t+1}^d} (1 - x_{t+1}) (V_{t+1}^N - V_{t+1}^U) \right] \frac{P_t^d}{P_t^c}$$

The gains induced by the matching friction generates a wedge between the maximum producer wage that the firm is willing to pay and the minimum producer wage that a worker is willing to accept:¹⁴

$$\begin{aligned} w_t^{UB} &= w_t + S_t^F = w_t + B \epsilon_t x_t^\vartheta \zeta_t^x \\ w_t^{LB} &= w_t - S_t^U \frac{P_t^c}{P_t^d} = \frac{\zeta_t^N A_L N_t^{\sigma_L}}{\psi_t^z} - \beta(1 - \delta) E_t \left[\frac{\psi_{t+1}^z}{\psi_t^z \mu_{t+1}^z \pi_{t+1}^d} (1 - x_{t+1}) (V_{t+1}^N - V_{t+1}^U) \right] \end{aligned}$$

The firm would be willing to pay up to a premium given by the hiring cost of an additional replacement worker. The worker would accept going down not just to the current marginal rate of substitution, but even further to accommodate the benefit of the probability of not being separated next period when one might not otherwise find another job. Any wage rate within this wage band could be an equilibrium wage. In particular, if the worker and firm have the same bargaining power, they would share the surplus equally $S_t^F = S_t^H \frac{P_t^c}{P_t^d}$, and thus a symmetric Nash-bargaining wage lies at the center of the wage band (for example, see Blanchard and Gali (2010)):

$$w_t^* = \frac{w_t^{UB} + w_t^{LB}}{2} = B \epsilon_t x_t^\vartheta \zeta_t^x + \frac{\zeta_t^N A_L N_t^{\sigma_L}}{\psi_t^z} - \beta(1 - \delta) E_t \left[\frac{\psi_{t+1}^z}{\psi_t^z \mu_{t+1}^z \pi_{t+1}^d} (1 - x_{t+1}) (B \epsilon_{t+1} x_{t+1}^\vartheta \zeta_{t+1}^x) \right] \quad (2.17)$$

Eq. (2.17) implies that when the hiring cost decreases, firms can fill a job more cheaply and are therefore less willing to pay a higher wage. Similarly, if workers' opportunity cost increases, they demand a higher wage. In addition, if the future expected gain increases, which may due to lower future opportunity costs, workers are less resistant to a wage cut today, and therefore the wage rate is lower. In the case of no hiring cost ($B = 0$), the Nash bargaining wage only depends on workers' marginal rate of substitution, and so $w_t^* = \frac{\zeta_t^N A_L N_t^{\sigma_L}}{\psi_t^z}$.

However, as argued by Shimer (2005), the real wage determined by Nash bargaining cannot generate the observed fluctuations of unemployment. The reason is that the real wage is determined by splitting the surplus with fixed portions between firm and household. As the upper and lower bounds of the wage band fluctuate due to the driving forces at the business cycle frequency, the real wage determined by Nash bargaining also fluctuates, often by a substantial amount. A volatile real wage scheme thus absorbs most of the driving forces, leaving unemployment relatively (too) stable. Recognizing the problem, Hall (2003); (2005) motivates a rigid real wage from wage norms, which insulates the real wage from movements of the upper and lower bounds insofar as it remains inside the bounds. Since the real wage lies inside the bounds, the real wage is consistent with equilibrium and there is no unrealized gain. However, this specification induces discrete movements in the real wage, which makes estimation hard. Instead, Hall (2003) shows that the aggregate real wage has an adaptive component if one were to allow for idiosyncratic random shifts in the bargaining set and then to aggregate over individual wage decisions. Alternatively, we can assume last period's wage is the wage norm for the current period. To further motivate the real wage following an adaptive process, we note that, in reality, firms and workers negotiate a wage bargain based on available information. Most likely, neither knows the actual value of unobserved technology parameters, particularly as they apply over the future life of the negotiated contract. Both may be able to make inferences from various signals received about productivity, but they will be aware of potential errors. These errors may lead them to take a more cautious approach

¹³ If labor does not have any monopoly power, the flexible wage should equal in equilibrium the marginal rate of substitution between consumption and work, as well as the net marginal product of labor.

¹⁴ We could rewrite the surpluses as $S_t^F = w_t^{UB} - w_t$ and $S_t^H = (w_t - w_t^{LB}) \frac{P_t^d}{P_t^c}$.

to real wage changes, adjusting them only partly in response to any received productivity signals, thus making the real wage history-dependent. To cope with these ideas, we use the following rule:

$$w_t = fw_{t-1} + (1-f)w_t^* \quad (2.18)$$

where f measures the degree of the real wage rigidity and $f = 0$ corresponds to the flexible Nash bargaining wage.¹⁵

2.3.2. Nominal wage rigidity

For this alternative, we assume there is a ‘firm’ that aggregates and transforms differentiated labor into a homogeneous labor service N_t , which can be used in the production of intermediate goods. The transformation process uses the following technology:

$$N_t = \left[\int_0^1 N_{j,t}^{\frac{1}{\lambda_t^w}} dj \right]^{\lambda_t^w}, \quad 1 \leq \lambda_t^w < \infty$$

where $N_{j,t}$ is monopolistic labor supply by household j at time t and λ_t^w is the wage markup that follows:

$$\lambda_t^w = (1 - \rho_{\lambda^w})\bar{\lambda}^w + \rho_{\lambda^w}\lambda_{t-1}^w + \epsilon_t^{\lambda^w}$$

where $\epsilon_t^{\lambda^w}$ is an i.i.d. shock. Profit maximization ensures the demand for labor j is:

$$N_{j,t} = \left(\frac{W_t}{W_{j,t}} \right)^{\frac{\lambda_t^w}{\lambda_t^w - 1}} N_t$$

where W_t is the nominal wage rate of the homogeneous labor service and $W_{j,t}$ is the nominal wage rate for household j . Since labor supply is differentiated, households have the power to set their nominal wage, but are subject to the Calvo (1983) type of nominal rigidity. In each period, household j faces a random probability ξ_w that she cannot change her nominal wage, and she partially indexes the wage according to:

$$W_{j,t} = (\pi_{t-1}^c)^{\kappa_w} (\pi_t^T)^{1-\kappa_w} \mu_t^z W_{j,t-1}$$

where π_t^c is the CPI inflation rate and κ_w is the degree of indexation¹⁶ (to prior inflation relative to the steady state inflation rate, which is the central bank’s target, π_t^T (discussed in the 2.4). With probability $1 - \xi_w$, household j receives a ‘price signal’ that she can re-optimize her wage at W_t^{new} . The aggregate wage rate is then:

$$W_t = \left[\xi_w \left((\pi_{t-1}^c)^{\kappa_w} (\pi_t^T)^{1-\kappa_w} \mu_t^z W_{t-1} \right)^{\frac{1}{1-\lambda_t^w}} + (1 - \xi_w) (W_t^{new})^{\frac{1}{1-\lambda_t^w}} \right]^{1-\lambda_t^w}$$

Once the nominal wage is set, households inelastically meet firms’ demand for labor. Stationarizing in real terms by dividing both sides by $P_t^d Z_t$, and then linearizing, gives:

$$\hat{w}_t = \xi_w (\hat{w}_{t-1} + \kappa_w \hat{\pi}_{t-1}^c + (1 - \kappa_w) \hat{\pi}_t^T - \hat{\pi}_t^d) + (1 - \xi_w) \hat{w}_t^{new}$$

The re-optimized real wage, w_t^{new} , hinges on the balance between the marginal utility of wage income and the marginal disutility from working, and applying this yields the equilibrium aggregate real wage rate:

$$\begin{aligned} \hat{w}_t = & \tau_b (\hat{w}_{t-1} + \kappa_w \hat{\pi}_{t-1}^c + (1 - \kappa_w) \hat{\pi}_t^T - \hat{\pi}_t^d) + \tau_c \left(\hat{\zeta}_t^N + \sigma_L \hat{N}_t - \hat{\psi}_t^z + \hat{\lambda}_t^w \right) \\ & + \tau_f \left(E_t [\hat{w}_{t+1}] - \kappa_w \hat{\pi}_t^c - (1 - \kappa_w) \rho_\pi \hat{\pi}_t^T + E_t [\hat{\pi}_{t+1}^d] \right) \end{aligned} \quad (2.19)$$

where $\tau_b \equiv \frac{(1+\tau)\xi_w}{1+\xi_w\tau+\beta\xi_w^2}$, $\tau_c \equiv \frac{(1-\beta\xi_w)(1-\xi_w)}{1+\xi_w\tau+\beta\xi_w^2}$, $\tau_f \equiv \frac{\beta\xi_w}{1+\xi_w\tau+\beta\xi_w^2}$, $\tau_b + \tau_f + \tau_c = 1$, and $\tau \equiv \frac{\bar{\lambda}^w \sigma_L (1-\beta\xi_w)}{\bar{\lambda}^w - 1}$. $\psi_t^z \equiv z_t P_t^d v_t$ is the stationarized marginal utility from income.

Eq. (2.19) shows the real wage is set as a weighted average of lagged and future expected wages and the current marginal rate of substitution. Unlike under real wage rigidity–recall (2.17)–hiring costs do not directly affect the real wage under nominal wage rigidity. This is because the wage is here being set by households, not by employers. Instead, hiring costs work indirectly by restricting labor demand, thus limiting the movement in N_t . Setting $\xi_w = 0$ gives the real wage without nominal wage rigidity:

$$\hat{w}_t = \hat{\zeta}_t^N + \sigma_L \hat{N}_t - \hat{\psi}_t^z + \hat{\lambda}_t^w$$

in which case the real wage only depends on the markup and the marginal rate of substitution between labor and consumption.

¹⁵ Gali and van Rens (2010), on the other hand, introduce an endogenous wage rigidity rule:

$$w_t = f_{t-1} w_{t-1} + (1 - f_{t-1}) w_t^*$$

where w_t^* is the symmetric Nash bargaining wage as discussed above, and f_{t-1} measures the degree of the endogenous real wage rigidity, that captures the idea that the wage is more likely to adjust when it approaches the boundaries of the bargaining set.

¹⁶ In Sheen and Wang (2014), we set κ_w to 1, which assumed perfect indexation to lagged inflation. This assumption unfairly penalized nominal wage rigidity when comparing its fit to the data with real wage rigidity.

2.4. Monetary policy

As in [Adolfson et al. \(2007\)](#), monetary policy follows:

$$\hat{R}_t = \rho_R \hat{R}_{t-1} + (1 - \rho_R)[\hat{\pi}_t^T + r_\pi(\hat{\pi}_{t-1}^c - \hat{\pi}_t^T) + r_y \hat{y}_{t-1} + r_s \hat{s}_{t-1}^r] + r_\Delta^\pi \Delta \hat{\pi}_t^c + r_\Delta^y \Delta \hat{y}_t + \epsilon_t^R$$

where $\hat{\pi}_t^c$ is the CPI inflation, $\hat{y}_t$ is the output gap and $\hat{s}_t^r$ denotes the log-linearized real exchange rate with $S_t^r = S_t^n \frac{P_t^*}{P_t^c}$. The parameter ρ_R measures the persistence in the interest rate. r_π , r_y and r_s measure the interest rate responses to deviations of lagged CPI inflation from the current inflation target $\hat{\pi}_t^T$, the lagged steady state deviations of output, $\hat{y}_{t-1}$ and of the real exchange rate, $\hat{s}_{t-1}^r$. The inflation target is assumed to follow an AR(1) process:

$$\hat{\pi}_t^T = \rho_\pi \hat{\pi}_{t-1}^T + \epsilon_t^T$$

The central bank is also assumed to respond to the current changes in the inflation and the output gap $\Delta \hat{\pi}_t^c$ and $\Delta \hat{y}_t$. Since the central bank sets the interest rate, it adjusts the outstanding money supply to match money demand by buying and selling domestic interest-earning assets. The government through the central bank issues all forms of money–cash and interest-earning assets—and continuously balances its budget with endogenous lump-sum taxes or transfers to households.

2.5. Goods market clearing

Since intermediate firms are inherently similar, they make identical decisions at equilibrium. Therefore (2.2) can be expressed as $Y_t = \epsilon_t z_t^{1-\alpha} K_t^\alpha N_t^{1-\alpha} - z_t \phi$. The aggregate resource constraint is given by:

$$C_t^d + I_t^d + X_t \leq Y_t - \tilde{a}(u_t) \tilde{K}_t - G_t H_t \quad (2.20)$$

so that net aggregate production accounts for the capital utilization cost $\tilde{a}(u_t)$ and the hiring cost $G_t H_t$. Substituting the demand for domestic consumption goods (2.10), domestic investment good (2.12), exports (2.7), and the hiring costs (2.3) into (2.20), and dividing both sides by z_t gives:

$$(1 - \omega_c) \left[\frac{P_t^d}{P_t^c} \right]^{-\eta_c} c_t + (1 - \omega_i) \left[\frac{P_t^d}{P_t^i} \right]^{-\eta_i} i_t + \left[\frac{P_t^x}{P_t^*} \right]^{-\eta_f} y_t^* \frac{z_t^*}{z_t} \leq y_t - \tilde{a}(u_t) \frac{\tilde{k}_t}{\mu_t^z} - g_t H_t$$

For simplicity, assume the foreign steady-state growth rate is the same as for the domestic economy. Define $\tilde{z}_t^* = \frac{z_t^*}{z_t}$ as a stationary shock that measures the degree of asymmetry in permanent shocks to technological progress between the foreign and the domestic economy. At the steady state, $\tilde{z}_t^* = 1$. Its log-linearized stochastic process is:

$$\hat{\tilde{z}}_{t+1}^* = \rho_{\tilde{z}^*} \hat{\tilde{z}}_t^* + \epsilon_{t+1}^{\tilde{z}^*}$$

Aggregate foreign assets must satisfy:

$$S_t^n B_{t+1}^* = S_t^n P_t^x X_t - S_t^n P_t^c (C_t^m + I_t^m) + R_{t-1}^* \Phi(a_{t-1}, \tilde{\phi}_{t-1}) S_t^n B_t^*$$

where $R_{t-1}^* \Phi(a_{t-1}, \tilde{\phi}_{t-1})$ represents the risk-adjusted gross interest rate and $a_t \equiv \frac{S_t^n B_{t+1}^*}{P_t^d z_t}$, which represents the stationarized real net foreign asset position in domestic value.

2.6. Loan market-clearing

Domestic loan market-clearing requires the borrowing in the form of the wage bill to equal the lending of the monetary system:

$$W_t N_t = \mu_t^m M_{t-1} - Q_t$$

where Q_t is nominal cash holding by households as defined in (2.8) and M_t is total domestic assets held including interest-earning bonds. μ_t^m is the gross monetary growth rate at time t :

$$\mu_t^m = \frac{M_t}{M_{t-1}}$$

Dividing both sides by z_t and P_t^d gives:

$$w_t N_t = \frac{\mu_t^m m_{t-1}}{\pi_t^d \mu_t^z} - q_t$$

Table 1
Calibrated parameters.

Parameter	Description	Value	Source
β	discount factor	0.997	matches sample average interest rate
δ	separation rate	0.1	Ponomareva and Sheen (2010)
ϑ	elasticity of hiring cost to labor market condition	1	Blanchard and Gali (2010)
δ_k	depreciation rate	0.013	Jääskelä and Nimark (2011)
α	capital share	0.29	Jääskelä and Nimark (2011)
ω_c	import share of consumption good	0.2	Jääskelä and Nimark (2011)
ω_i	import share of investment good	0.5	Jääskelä and Nimark (2011)
σ_q	money demand curvature parameter	10.62	Adolfson et al. (2007)
$\bar{\lambda}^w$	steady-state labor markup	1.05	Adolfson et al. (2007)
σ_L	labor supply elasticity	1	Adolfson et al. (2007)
A_L	labor disutility constant	1	
Aq	money utility constant	0.3776	Adolfson et al. (2007)
$\bar{U}$	steady-state level of unemployment	0.065	sample average unemployment rate
$\bar{\pi}^d$	steady-state level of domestic goods inflation	1.0065	Jääskelä and Nimark (2011)
ρ_{λ^d}	persistence of final domestic goods markup	0.5	
ρ_π	persistence of perceived inflation	0.5	Adolfson et al. (2007)
$\frac{a''(1)}{a'(1)}$	steady-state utilization cost parameter	0.049	Adolfson et al. (2007)

2.7. The foreign economy

The foreign economy is summarized by three variables—inflation π_t^* , the output gap $\hat{y}_t^*$ and the short-term interest rate $\hat{R}_t^*$.¹⁷ They are modelled as a VAR(1) process: $(F_t^* \equiv [\hat{y}_t^* \ \pi_t^* \ \hat{R}_t^*]')$

$$F_t^* = \rho_F^* F_{t-1}^* + \epsilon_t^*$$

3. Model selection

We use quarterly Australian data from 1993Q2 to 2013Q2, including CPI inflation, the GDP deflator, nominal cash rate, real GDP, consumption, investment, imports and exports, G7-weighted real exchange rate, unemployment rate and real wages, G7 real GDP, average inflation and nominal cash rate (Appendix A lists the sources of the data). The starting date was chosen because it begins when the Reserve Bank of Australia commenced inflation targeting. The CPI inflation is measured as the quarterly percentage growth of the CPI price index. Real GDP, consumption, investment, import and export are chain-volume measures, and the real wage is measured by nominal average weekly earnings per adult divided by the GDP deflator. The estimation maps Θ in the theoretical model to the data, where:

$$\Theta \in \{\hat{\pi}_t^c, \hat{\pi}_t^d, \hat{w}_t, \hat{y}_t, \hat{c}_t, \hat{l}_t, \hat{X}_t, \hat{M}_t^m, \hat{R}_t, \hat{s}_t^r, \hat{U}_t, \hat{R}_t^*, \hat{y}_t^*, \hat{\pi}_t^*\}$$

where a variable with a caret (^) represents its log-linearized counterpart. We map the growth rates of output (and its components) and real wages with variables in the theoretical model, so that we can identify the common trend. Appendix C presents the mapping of the variables in Θ to the observed data series.

3.1. Calibration and priors

We calibrate a number of parameters, which are shown in Table 1. $\beta = 0.997$ implies the steady-state nominal interest rate is 5.4%, which matches the average interest rate in the sample period. Setting δ at 0.1 delivers a steady-state monthly separation rate of 3.45% and a job-finding rate of 25%, which is consistent with the Australian labor market in the period under discussion (see Ponomareva and Sheen (2010) for details).¹⁸ As in Blanchard and Gali (2010), we set ϑ at 1, which implies the hiring cost is unit-elastic to current labor market conditions. We follow Jääskelä and Nimark (2011), in calibrating for the Australian economy by setting the depreciation rate δ_k at 0.013, the labor share of production α at 0.29, and the share of consumption and investment in imports, ω_c and ω_i at 0.2 and 0.5. Following Adolfson et al. (2007), we set the curvature of the money demand function σ_q at 10.62, the elasticity of labor supply σ_L at 1, and the constants that determine the level of utility from real balances Aq at 0.337. Since the steady-state markup for labor-producing firms $\bar{\lambda}^w$ is not jointly identified with the labor supply shock, we set the value at 1.05. We use the sample average for the steady-state level of the annual unemployment rate $\bar{U}$ (calibrated at 6.5%) and for the inflation rate of domestic production (2.6%). The utility parameter A_L is set to 1, which has implications for the steady state level of the marginal disutility of labor and thus the real wage, and then the capital to labor ratio.

¹⁷ The foreign interest rate is nonstationary for the sample period we consider. Therefore we apply an HP filter to this interest rate.

¹⁸ We calculate the implied monthly rate from the quarterly rate by solving $m + (1 - m)m + (1 - m)^2m = q$, where $q \in \{\delta, \bar{x}\}$. The steady-state job finding rate $\bar{x}$ is calculated based on a 7% steady-state unemployment rate.

Table 2
Comparison of models using the Bayes Factor.

	Log-Likelihood	BF(Null: RWR)	BF(Null: NHC)
Benchmark models			
NWR, HC	–1270.07	16.99	3.19
NWR, NHC	–1273.26	12.39	—
RWR, HC	–1287.06	—	–1.41
RWR, NHC	–1285.65	—	—
Full backward indexation models			
NWR, HC	–1310.26	6.75	8.54
NWR, NHC	–1318.80	1.60	—
RWR, HC	–1317.01	—	3.39
RWR, NHC	–1320.39	—	—
Without observations on GDP deflator π_t^d			
NWR, HC	–1148.81	–4.91	–1.64
NWR, NHC	–1147.18	–5.57	—
RWR, HC	–1143.91	—	–2.30
RWR, NHC	–1141.61	—	—
Models using data prior to the financial crisis of 2008			
NWR, HC	–858.43	16.36	2.77
NWR, NHC	–861.20	12.33	—
RWR, HC	–874.79	—	–1.26
RWR, NHC	–873.53	—	—
NWR: Nominal wage rigidity; HC: Hiring cost BF: Log of the Bayes factor			
RWR: Real wage rigidity NHC: No hiring cost			

Note: A value of the Bayes Factor (BF) between 1 and 3 indicates positive evidence against the null model H_0 , 3 and 5 indicates strong evidence, and greater than 5 indicates very strong evidence (Kass and Raftery (1995)).

Priors

For the key parameter B that determines the steady-state level of hiring cost, we choose a an inverse gamma distribution with mean 0.4 and standard deviation of 10. Conditional on the prior for other parameters, this choice implies a prior distribution for the hiring cost with a mean of 1.1% and a mode of 0.4%. We follow Adolfson et al. (2007) and Jääskelä and Nimark (2011), in choosing priors for the rest of the model. For parameters that ought to be bounded between 0 and 1, such as the persistence parameters, we use a beta distribution. For elasticities, steady-state markups and standard deviation of shocks, we expect the mass of the distribution to be concentrated at small values, but do not rule out possible large values; thus inverse gamma distributions are used. Some of the shocks appear in the model scaled by a non-linear function of deep parameters. We set the same prior for the ‘reduced’ form of these shocks as for unscaled shocks—with a mean 0.5 and standard deviation 2. For the rest, we simply employ normal distributions. Since we were unable to identify the hiring cost shock, ϵ_t^x , the imported investment good markup shock, $\epsilon_t^{\lambda^m}$, and the nominal wage markup shock, $\epsilon_t^{\lambda^w}$, we set their standard errors to zero.¹⁹ Table 3 shows our particular choice of priors for the model that we prefer, as determined in the next section.

3.2. Which labor market frictions?

We begin by presenting our results that lead us to prefer the model with hiring costs and nominal wage rigidity. Later we provide detailed estimates of our preferred model, including impulse responses and variance decompositions²⁰.

Four alternative models are investigated—with or without hiring costs, and with nominal or real wage rigidity.

The top panel of Table 2 gives results for the benchmark models and shows that nominal wage rigidity is preferred to real wage rigidity.

Comparing posterior log-likelihoods, the benchmark model with hiring costs and nominal wage rigidity fits the data the best, with a value of –1270.07. Using the log of the Bayes factor as the criterion, models with nominal wage rigidity are very strongly favored by the data regardless of the existence of hiring costs²¹—assuming a null hypothesis of real wage rigidity, the log of the Bayes factor is 16.99 (–1270.07–(–1287.06)) with hiring costs and 12.39 (–1273.26–(–1285.65)) without hiring

¹⁹ From Eq. (2.19) for the real wage under nominal wage rigidity, the wage markup shock λ_t^w cannot be jointly identified with the labour supply shock ζ_t^N . In order to make it comparable with the real wage rigidity models, we choose to estimate ζ_t^N and fix λ_t^w .

²⁰ The models are solved numerically using Dynare. The Kalman filter is employed to evaluate the model likelihood, and the Markov Chain-Monte Carlo (MCMC) method is used to conduct the posterior simulations. Half a million draws are taken, of which 30% are discarded to minimize the impact of initial values. We also control the variance of the candidate distribution from which the simulation draws to achieve an acceptance rate around 40%.

²¹ Kass and Raftery (1995) show the Bayes factor is equivalent to the posterior odds ratio if one places an equal prior weight on the two competing models. Denote $B_{10} = \frac{p(\text{Data}|H_1)}{p(\text{Data}|H_0)}$ as the Bayes factor, where $p(\text{Data}|H_0)$ is the likelihood of data conditional on the null hypothesis H_0 and $p(\text{Data}|H_1)$ is the likelihood conditional on the alternative model H_1 . They suggest a value of $\ln B_{10}$ between 1 and 3 indicates positive evidence against model H_0 , 3 and 5 indicates strong evidence, whereas a value greater than 5 indicates very strong evidence in favor of that model.

Table 3

Priors and posteriors for the benchmark model (nominal wage rigidity and hiring costs).

Parameter	Description	Priors			Posterior			
		Distribution	Mean	Std	Mean	Std	5%	95%
b	habit formation	Beta	0.65	0.10	0.65	0.06	0.55	0.75
B	tightness coefficient	InvGamma	0.40	10.0	0.21	0.05	0.11	0.30
$\tilde{S}'$	curvature of investment adjustment cost	Normal	7.69	2.50	6.39	2.15	2.97	9.59
$\tilde{\phi}_a$	risk premium	InvGamma	0.01	0.002	0.010	0.002	0.007	0.012
η_c	imported consumption elasticity	InvGamma	1.50	4.00	1.15	0.03	1.10	1.20
η_i	imported investment elasticity	InvGamma	1.50	4.00	2.30	0.30	1.79	2.81
η_f	export elasticity	InvGamma	1.50	4.00	1.33	0.09	1.12	1.57
$\tilde{\mu}^z$	steady-state growth rate	Normal	1.004	0.002	1.004	0.001	1.003	1.005
$\tilde{\lambda}^d$	steady-state markup: domestic	InvGamma	1.20	2.00	2.56	0.46	1.83	3.25
$\tilde{\lambda}^{mc}$	steady-state markup: imported-consumption	InvGamma	1.20	2.00	1.96	0.32	1.36	2.60
$\tilde{\lambda}^{mi}$	steady-state markup: imported-investment	InvGamma	1.20	2.00	1.19	0.07	1.07	1.30
$\tilde{\lambda}^w$	steady-state markup: wage rate	InvGamma	1.20	2.00	1.04	0.01	1.01	1.08
Calvo lottery and backward indexation								
ξ_d	domestic firm	Beta	0.675	0.10	0.53	0.13	0.36	0.71
ξ_c^m	consumption import firm	Beta	0.675	0.10	0.91	0.02	0.87	0.95
ξ_i^m	investment import firm	Beta	0.675	0.10	0.26	0.05	0.19	0.35
ξ_x	exporter	Beta	0.675	0.10	0.89	0.03	0.83	0.95
ξ^w	wage setter	Beta	0.675	0.10	0.66	0.12	0.50	0.83
κ_d	backward indexation: domestic	Beta	0.50	0.20	0.33	0.13	0.08	0.57
κ^{mc}	backward indexation: imported-consumption	Beta	0.50	0.20	0.12	0.06	0.02	0.22
κ^{mi}	backward indexation: imported-investment	Beta	0.50	0.20	0.35	0.19	0.07	0.61
κ_x	backward indexation: export	Beta	0.50	0.20	0.12	0.06	0.02	0.21
κ_w	backward indexation: wage	Beta	0.50	0.20	0.33	0.13	0.10	0.54
Monetary policy parameters								
ρ_R	interest rate smoothing	Beta	0.80	0.05	0.86	0.02	0.82	0.90
r_π	inflation response	Normal	1.75	0.30	2.11	0.25	1.71	2.51
r_y	output response	Normal	0.125	0.05	0.06	0.03	0.01	0.11
r_s	real exchange rate response	Normal	0.00	0.05	−0.01	0.02	−0.04	0.01
$r_{\Delta\pi}$	inflation change response	Normal	0.00	0.10	0.17	0.04	0.10	0.24
$r_{\Delta y}$	output change response	Normal	0.00	0.10	0.03	0.02	−0.01	0.07
Persistence parameters								
ρ_{ζ^c}	consumption preference	Beta	0.85	0.10	0.81	0.10	0.67	0.96
ρ_{ζ^N}	labor preference	Beta	0.85	0.10	0.69	0.12	0.50	0.86
ρ_{μ^z}	permanent technology	Beta	0.85	0.10	0.76	0.07	0.65	0.88
ρ_ϵ	temporary technology	Beta	0.85	0.10	0.93	0.02	0.88	0.99
ρ_ϕ	risk premium	Beta	0.85	0.10	0.96	0.02	0.92	0.99
ρ_Γ	investment-specific technology	Beta	0.85	0.10	0.36	0.09	0.22	0.51
$\rho_{\tilde{z}^*}$	asymmetric foreign technology	Beta	0.85	0.10	0.89	0.06	0.79	0.99
ρ_{λ^x}	markup: export	Beta	0.50	0.20	0.14	0.07	0.02	0.25
Standard errors of exogenous shocks								
σ_{ζ^c}	preference shock: consumption	InvGamma	1.69	6.74	1.68	0.34	1.14	2.22
σ_{ζ^N}	preference shock: labor supply	InvGamma	28.6	114.3	19.31	3.98	12.70	25.60
σ_{μ^z}	permanent technology shock	InvGamma	0.50	2.00	0.23	0.04	0.16	0.31
σ_ϵ	temporary technology shock	InvGamma	0.50	2.00	0.38	0.09	0.24	0.51
σ_ϕ	risk premium shock	InvGamma	0.50	2.00	0.27	0.06	0.17	0.38
σ_Γ	investment specific technology shock	InvGamma	0.03	0.12	0.09	0.01	0.07	0.11
$\sigma_{\tilde{z}^*}$	asymmetric foreign technology shock	InvGamma	0.50	2.00	0.30	0.06	0.21	0.39
σ_R	monetary policy shock	InvGamma	0.50	2.00	0.12	0.01	0.10	0.15
σ_{λ^d}	markup shock: domestic production	InvGamma	4.76	18.9	1.11	0.20	0.63	1.55
$\sigma_{\lambda^{cm}}$	markup shock: imported-consumption	InvGamma	4.76	18.9	173.5	20.8	137.7	209.5
σ_{λ^x}	markup shock: export	InvGamma	4.76	18.9	122.7	17.3	89.9	156.1
σ_{π^T}	inflation target	InvGamma	0.05	2.00	0.05	0.01	0.01	0.08

costs. Comparing models with and without hiring costs, the data suggest a strong evidence in favor of the existence of the hiring costs under nominal wage rigidity but there is some evidence against hiring costs under (our non-preferred) real wage rigidity.

The estimates of the structural parameters depend on the assumptions of the existence of hiring costs and the type of wage rigidity. Figs. 1 and 2 show the effects on selected parameters.²² For details of the estimation results, see Appendix E.

The role of hiring costs

The introduction of hiring costs increases labor costs and thus dampens firms' demand for labor, allowing a role for hiring costs to drive inflation through real marginal cost (see Eq. (2.4) and (2.6)). Since hiring costs are persistent (through

²² Appendix E plots the posterior distributions for all parameters.

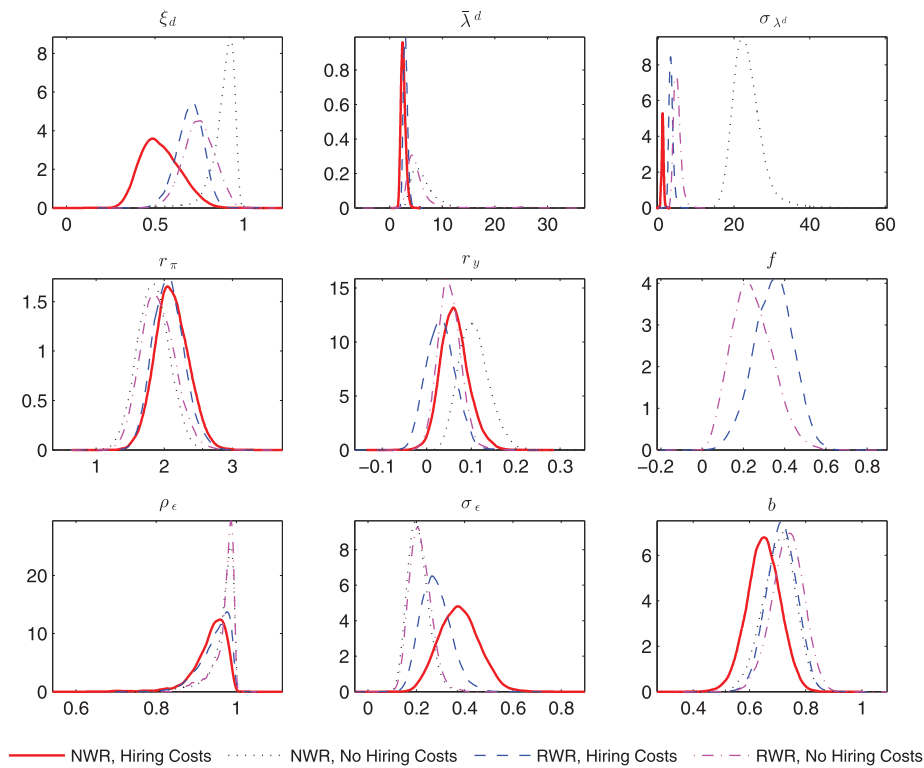


Fig. 1. Comparison of posterior distributions: effects of the existence of hiring costs.

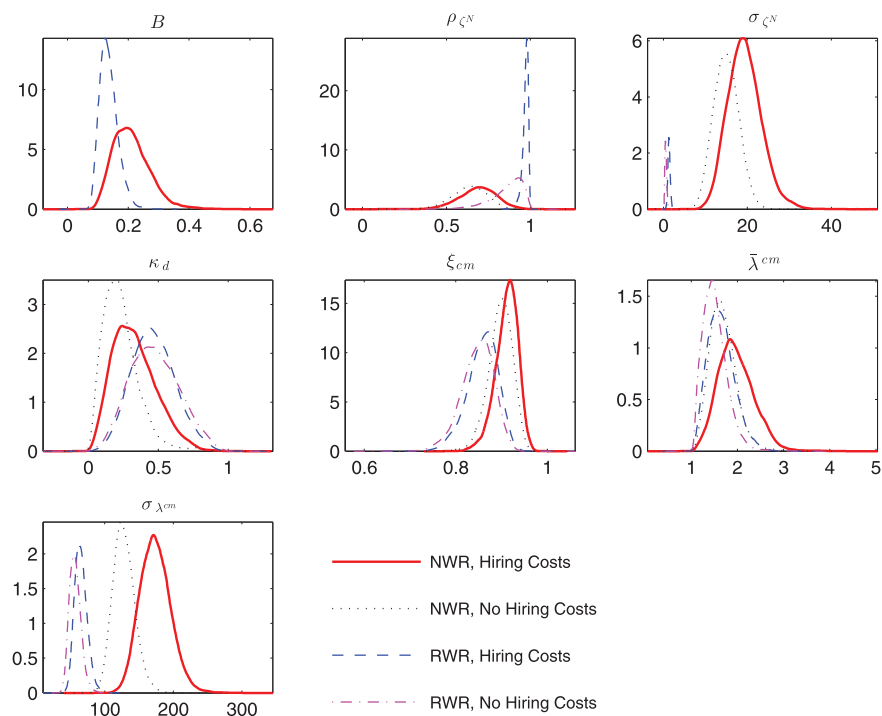


Fig. 2. Comparison of posterior distributions: effects of types of wage rigidity.

temporary technology shocks), models with hiring costs require a smaller Calvo parameter for domestic intermediary producers ξ_d (and therefore, a larger weight put on the real marginal cost) to match the inflation persistence in the data. The first row of Fig. 1 shows the impact on parameters related to domestic intermediate producers. The point estimate for the Calvo lottery ξ_d is 0.53 (0.70) compared to 0.88(0.74) under nominal(real) wage rigidity. A more flexible contract implies a lower steady-state markup (λ^d) and markup volatility (σ_{λ^d}). Since the implied prices are more flexible, the central bank places a larger weight on inflation and a smaller weight on the output gap, which is seen in the second row of Fig. 1. Under Nash bargaining, the hiring cost makes firms and workers consider both current and future expected hiring costs in the bargaining process, which leads to the Nash-bargaining wage w_t^* being directly indexed to the temporary technology shock. The inclusion of this shock introduces additional volatility into the real wage, thus requiring a higher persistence parameter f to match the persistence observed in the data. The point estimate increases from 0.24 to 0.34.

The last row of Fig. 1 shows the impact on the persistent parameter and the standard deviation of the temporary technology shock (ρ_ϵ and σ_ϵ) and the consumption habit formation parameter b . The hiring costs restrict labor demand, thus requiring a bigger and less persistent technology shock to shift labor demand and induce further fluctuations in the labor market. The point estimate of the persistence parameter of the temporary technology shock decreases from 0.97 (0.97) to 0.93 (0.95), and the standard deviation of the shock increases from 0.21(0.22) to 0.38 (0.27) under nominal (real) wage rigidity. Since hiring costs increase the persistence in real wages, they in turn increase the persistence in the marginal rate of substitution and thus aggregate consumption. Therefore, less habit formation is needed to force a match of the observed persistence in aggregate consumption. The habit formation parameter b reduces from 0.70 (0.73) to 0.65 (0.71) under nominal (real) wage rigidity. Although hiring costs reduce the amount of habit formation, all models still require a substantial amount of the latter to match the persistence of consumption.

The role of wage rigidity

As shown in Section 2.3, nominal wage rigidity fixes the nominal wage via Calvo-pricing and backward wage indexation while real wage rigidity dampens the real wage responses by placing some weight on the lagged real wage rate. Under real wage rigidity, hiring costs play a substantial role in driving real wages. On the other hand, the impact of hiring costs is muted under nominal wage rigidity, but allows CPI inflation (and the GDP deflator) to play a larger role in driving the dynamics of the real wage.

The first row of Fig. 2 shows the impact of wage rigidities on the estimates of the hiring cost parameter B and the persistence (ρ_{ζ^N}) and standard deviation (σ_{ζ^N}) of the labor preference shock ζ^N . Comparing models with hiring costs, real wage rigidity models allow an additional channel for hiring costs to impact on the economy, resulting in a smaller estimate of the hiring cost parameter B . The point estimate increases from 0.14 to 0.21 under nominal wage rigidity, which implies an increase of the steady-state hiring costs to GDP ratio from 1.26 to 1.58%.²³ Both wage rigidity assumptions allow a positive contribution of the marginal disutility to real wage, independent of the existence of hiring costs. Note that the weight placed on this contribution (τ_2 and τ_c) depends on the respective wage assumptions. Our estimation results suggest nominal wage rigidity places a much smaller weight on marginal disutilities than real wage rigidity, thus requiring a larger and less persistent labor supply shock to match the moments in labor market variables.²⁴

The second and third row of Fig. 2 show the impact of wage rigidity on estimated parameters related to domestic and consumption-import firms. Nominal wage rigidity allows a role for domestic inflation in driving real wages, and assigns a large value to the persistence parameter τ_b .²⁵ Since real wages partially drive domestic inflation π_t^d , this implies that past inflation feed through past real wages in determining the current real wage, thus playing a role in driving current inflation. (see Eqs. (2.19), (2.4) and (2.6)) This additional channel of persistence for domestic price inflation crowds out the impact of backward indexation under nominal wage rigidity, which results in a lower estimate of the parameter κ_d . This lower persistence for domestic price inflation in turn requires a higher price persistence for consumption-importing firms to match the persistence in observed CPI inflation. The estimate increases from 0.86(0.85) to 0.91(0.90) for models with(without) hiring costs. A more rigid wage contract implies a higher steady-state markup for consumption-importing firms and a larger markup shock to match the observed moments for CPI inflation.

3.3. Robustness checks: Other model specifications and data mapping

The lower panels of Table 2 provide some additional information to support our preference of nominal wage rigidity with hiring costs.

The second panel shows the results if full indexation of wages and all prices is assumed across models. These include the Calvo parameters for domestic producers, importers, exporters and wage setters under nominal wage rigidity. Comparing the likelihood for each of the four variants, models with partial indexation perform significantly better than full indexation. Consider wage rigidity. The log of the Bayes factors are significantly weakened regardless of the existence of hiring costs, making the preference of nominal wage rigidity marginal under no hiring costs. This is because full indexation imposes too

²³ The steady-state hiring cost is calculated as $\frac{\bar{g}\bar{N}}{\bar{y}-\bar{g}\bar{N}}$.

²⁴ The estimation of the posterior mode implies that the real wage rigidity model attributes a large contribution to fluctuations in labor supply shocks compared to nominal wage rigidity model ($\tau_2 = 0.90$ and $\tau_c = 0.02$). In fact, the posterior distribution of the standard deviations of the reduced-form(unscaled) labor supply shock looks similar for the four variants we considered here.

²⁵ τ_b is 0.89 for our preferred model.

strong assumption on wage (and price) setting under nominal wage rigidity when a change signal is not received, making an biased competition between the two types of wage rigidity. On the other hand, full indexation makes models with hiring costs strongly preferred regardless of wage rigidity.²⁶

The third panel of Table 2 compares models when we omit data on the GDP price deflator π_t^d from the measurement equation, forcing the deflator to be endogenously determined in the model. This overturns the conclusion on the preference of nominal wage rigidity; instead, models with real wage rigidity are preferred regardless of the existence of hiring costs. This is because it unnecessarily restricts the indexation target, which is the deflator. This confirms the findings in Guerron-Quintana (2010), that the data choice has a big impact on the parameter estimation. In addition, he finds mapping more observables is better as long as it is accompanied by measurement. We therefore prefer to include the GDP deflator data in our estimation.

The last panel shows the results when we use data prior to the recent financial crisis, that is before 2007Q2. The results are qualitatively the same, with the preferred model being the model with nominal wage rigidity and hiring costs.

4. Detailed estimates of our preferred model: Nominal wage rigidity and hiring costs

Having established above the basis for our preferred model with nominal wage rigidity and hiring costs, its prior and posterior estimates are displayed in Table 3 and Appendix D shows the prior and posterior distributions. A key parameter in our setup of this model is B , which determines the steady-state level of hiring costs. We find B is sharply estimated with a mean of 0.21 and a 90% credibility interval bounded between 0.11 and 0.30. This implies the point estimate of the hiring costs to GDP ratio evaluated at the steady state is 1.58%, with a 90% credibility interval ranging from 0.85 to 2.26. Our estimation suggests statistically significant hiring costs in the Australian economy, and is consistent with the ABS estimates of 1.3% for the value-added industry for the year 2001–2002 (Australian Bureau of Statistics (2003)).²⁷

The point estimate of the habit formation parameter b is 0.65, indicating households prefer a relatively smooth consumption path. The investment adjustment cost $\bar{S}^I$ is 6.39. This implies the temporary elasticity of investment with respect to the current price of existing capital is 0.16, while the permanent elasticity is 51.75.

Our estimate of the risk premium parameter $\tilde{\phi}_a$ is 0.01, which indicates a 1% increase in net foreign assets reduces the domestic rate of interest (*ceteris paribus*) by 0.01%. Although this risk premium parameter is small in absolute value, it is statistically significant. The elasticities of substitution of imported consumption and investment to domestic-produced goods are 1.15 and 2.30 respectively, which indicates the demand for imported investment goods is substantially more sensitive to changes in relative price than imported consumption goods. The mean estimate for $\tilde{\mu}^z$ implies an annual real GDP growth rate of 1.6%, with a 90% confidence interval ranging from 1.2 to 2%. This is consistent with the sample average growth rate, which is 1.65%. The point estimates of steady-state markups show that domestic firms have the most, and investment-importing firms have the least market power compared to other types of firms. The estimate of the steady-state markup in the labour market ($\bar{\lambda}^w$) shows that labour unions have limited market power.

The point estimate of the Calvo lottery parameter for domestic firms is 0.53, which implies an average price fix duration of 2 quarters.²⁸ The Calvo parameter for consumption-importing and investment-importing firms are estimated at 0.91 (11 quarters) and 0.26 (1.4 quarters) respectively. The Calvo parameter for exporters is as high as for consumption-importing firms with the mean centered at 0.89 (9.1 quarters), which suggests that exporters and consumption-importers write longer-term contracts than other types of firms. The Calvo parameter for labor unions is evidently not identified in our preferred model. The estimation of the backward indexation parameters strongly suggests less than full indexation for all types of firms, with the estimated values ranging from 0.12 for consumption-importing and exporting firms to 0.35 for investment-importing firms.

The estimates of the interest rate rule suggest that the Reserve Bank of Australia (RBA) adjusted the nominal interest rate in response to inflation more than proportionately so as to influence the real interest rate (a policy requirement emphasized by Clarida et al. (1999); 2000)). The point estimate of the interest persistence parameter is 0.86, which indicates a substantial amount of interest rate smoothing. This may reflect the desire to affect the long-run interest rate (Goodfriend (1991); Woodford (1999)), or may approximate for a more complicated optimal monetary policy (Levin et al. (1999)), or may arise because monetary policy is conducted under uncertainty (Sack and Wieland (2000)). The point estimates also suggest the RBA placed little weight on the real exchange rate and output directly. Inflation change had a modest but relevant effect, but output change had little.

Our results indicate high persistence for shocks to temporary technology (ρ_ϵ), the risk premium on foreign bonds (ρ_ϕ) and asymmetric technology ($\tilde{z}^*$), with point estimates between 0.89 and 0.96. The consumption preference shock $\rho_{\zeta c}$, labor

²⁶ The choices of the data we map and the assumption on backward indexation led us to different conclusions in an earlier version of this research (in Sheen and Wang (2014)), where we assumed full backward indexation for prices and wage setters and omitted the GDP deflator as one of the observable series. There we found real wage rigidity model with hiring costs was the most preferred. The full indexation assumption, in particular for wage setters, made the comparison between nominal and real wage rigidity models biased towards the latter because it unnecessarily restricted the lag dependence of the wages on inflation to 1. Full backward indexation led to much stronger support for hiring costs. In addition, exclusion of the GDP deflator implied a poorer information set, biasing against nominal wage rigidity models.

²⁷ According to the Australian Bureau of Statistics (2003), the total industry value-added by employment replacement firms was \$8866.7 million, which accounted for 1.3% of GDP in 2001–2.

²⁸ If ξ as the probability that a firm cannot change its price at a given period, Calvo pricing implies the average price fix duration is $\frac{1}{1-\xi}$.

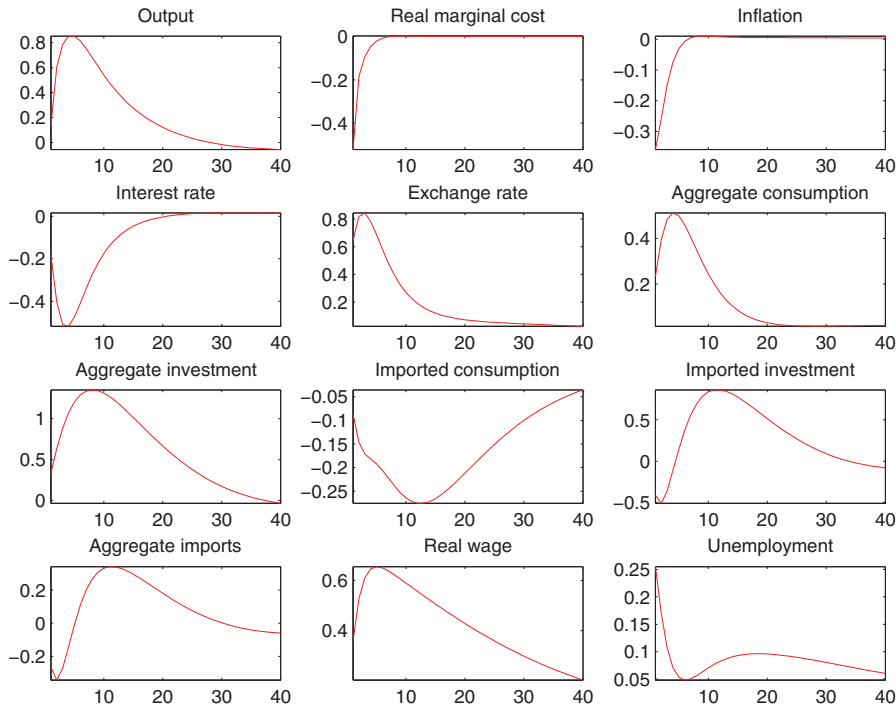


Fig. 3. Impulse responses: 1 standard deviation temporary technology shock.

supply preference shock ($\rho_{\xi N}$) and the permanent technology shock ρ_{μ^z} are less persistent (0.81, 0.69 and 0.76 respectively). In sharp contrast, the investment-specific technology (ρ_{Γ}) and export markup (ρ_{λ^x}) shocks have low persistence (0.36 and 0.14 respectively). Three of the estimated means of the standard deviations are large— $\sigma_{\xi N}$, $\sigma_{\lambda^{cm}}$ and σ_{λ^x} —simply because they appear in the model scaled by a small estimated coefficient that is a non-linear function of deep parameters.

4.1. Impulse responses

4.1.1. A positive temporary technology shock

Fig. 3 shows the impulse responses to a positive one-standard-deviation temporary shock to technology ϵ_t^ϵ for our preferred model with nominal wage rigidity and hiring costs. Following the shock, output expands and peaks after 5 quarters. The real marginal cost mc^d decreases, which drives down the domestic producers' inflation π^d . This allows the central bank to decrease the interest rate R , which causes the exchange rate S^N to depreciate.

Although both aggregate consumption c and investment i rise following the positive temporary technology shock, the shock decreases imported-consumption c^m and imported-investment i^m due to the exchange rate depreciation. This is because the income effects out-weigh the substitution effects as seen in Eqs. (2.11) and (2.13).

Following the shock, productivity increases and thus real wages rise. However, the increase in aggregate demand is restricted by the various rigidities present in the economy: hiring costs that add a premium to employment change; nominal price rigidities that constrain the decrease in price thus limiting the aggregate demand change; habit formation that dampens the increase in aggregate consumption; costly installation of investment goods and variable capital utilization that limit the increase in aggregate investment; and incomplete pass-through of export prices that restricts the increase in export demand. Therefore, the combination of rising wages and restricted aggregate demand reduces the demand for labor, and thus raises unemployment. Although output and unemployment move on impact in the same direction with this shock, we will show later that this apparent contravention of Okun's law is not a prediction of the overall model.

4.1.2. A monetary policy contraction shock

Fig. 4 shows the impulse responses for our preferred model after a positive one-standard-deviation shock to the interest rate. Following the shock, the interest rate rises, which reduces aggregate output, consumption and investment. The reduction in production cost decreases the real marginal cost, which in turn lowers inflation. The increase in the interest rate makes the domestic bond more attractive, which results in an appreciation of the domestic currency. This appreciation of the local currency makes importing firms more likely to substitute local products with foreign products. However, the response of imports depends on the substitution and income effects. Income effects dominate the response of imported consumption while substitution dominates for imported investment. Since the nominal wage is fixed initially and there is

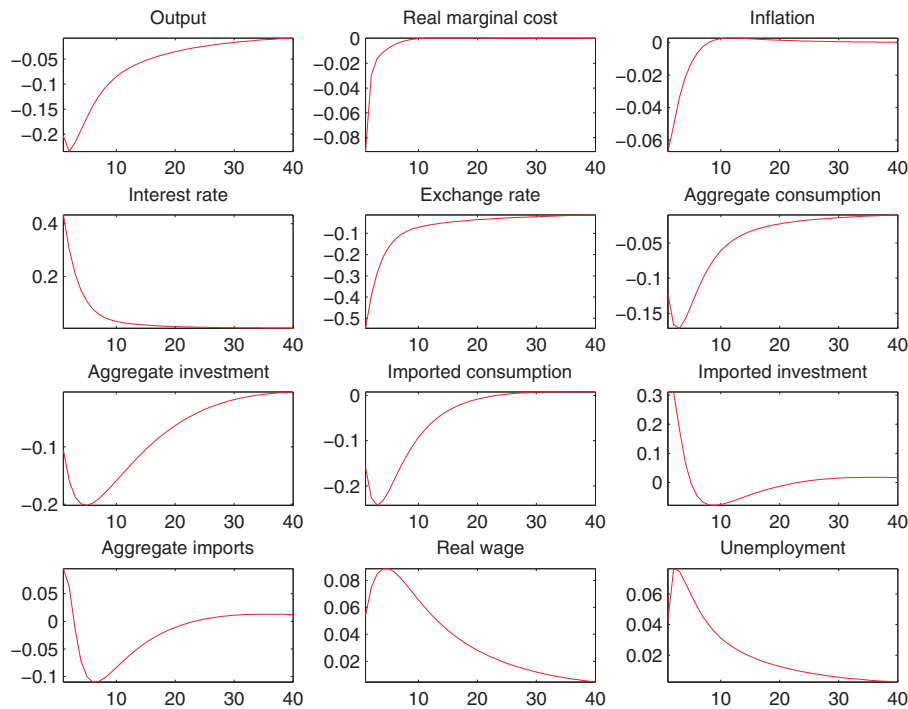


Fig. 4. Impulse responses: 1 standard deviation monetary policy shock.

a fall in domestic price inflation, the real wage rises after the shock. The combination of lower demand and higher wages raises the unemployment rate. Since output falls and unemployment rises, this shock is consistent with Okun's law.

4.2. Variance decomposition for the preferred model

Fig. 5 shows the forecast variance decomposition both on impact and in the long run of the various shocks on key endogenous variables. We aggregate the effects of both temporary and permanent technology shocks into one single measure (defined as Tech), and aggregate over all shocks related to the small open economy (SOE). The other shocks relate to consumption preferences (Pref-C), labor supply preferences (Pref-L), investment (Inv), monetary policy (MP) and the domestic markup (Markup-Dom). There are a number of notable results.

About 57% of the variance of unemployment ($\hat{U}$) on impact is explained by technology shocks (almost all temporary), which are reflected in variations of labor demand and thus hiring costs ($\hat{g}$). In the long run equilibrium, this contribution declines to about 46%, with the contribution of labor supply preference shocks jumping to 33%. Domestic markup shocks dominates the real wages in the short run (58%), whereas in the long run, technology shocks account for 70% of real wage fluctuations. labour preference shocks are moderate under both short run (20%) and long run (16%).

But, interestingly, technology shocks hardly play any role in driving short run output ($\hat{y}$) fluctuations, leaving aggregate demand-side factors to account for 72% (SOE: 32%, Pref-C: 11%, Inv: 29%) of output variance. In the long run however, technology and labor preference shocks on the aggregate supply side account for 68% (Tech: 34%, Pref-L: 34%) of output variations. Our results thus support the predictions by models from the new Keynesian tradition, which attribute short run output fluctuations to demand shocks and long run fluctuations to supply shocks.

Taking into account the effects of all shocks on output and unemployment, we find that estimated correlation between these two variables implied by the estimated model is -0.40 , which is evidence that the model is consistent with Okun's law.

Investment shocks contributed significantly in the short run to investment ($\hat{i}$) variation, and to a much lesser extent to output ($\hat{y}$, 29%), but have limited impact in the long run. Consumption preference shocks have a significant effect on consumption ($\hat{c}$) variation in the short run (43%), but play a lesser role in the long run. Taking into account all shocks, we find significant positive co-movement of the key domestic components of aggregate demand (with the correlation of consumption and output being 0.54 and investment and output being 0.43).

External shocks in this small open economy have important effects on inflation and consumption variance in both the short and long run. This channel operates largely through exchange rate variance. In the long run, the variance of investment and interest rates are dominated by external shocks. There are minimal impacts of external shocks on the labour market in the short and long run.

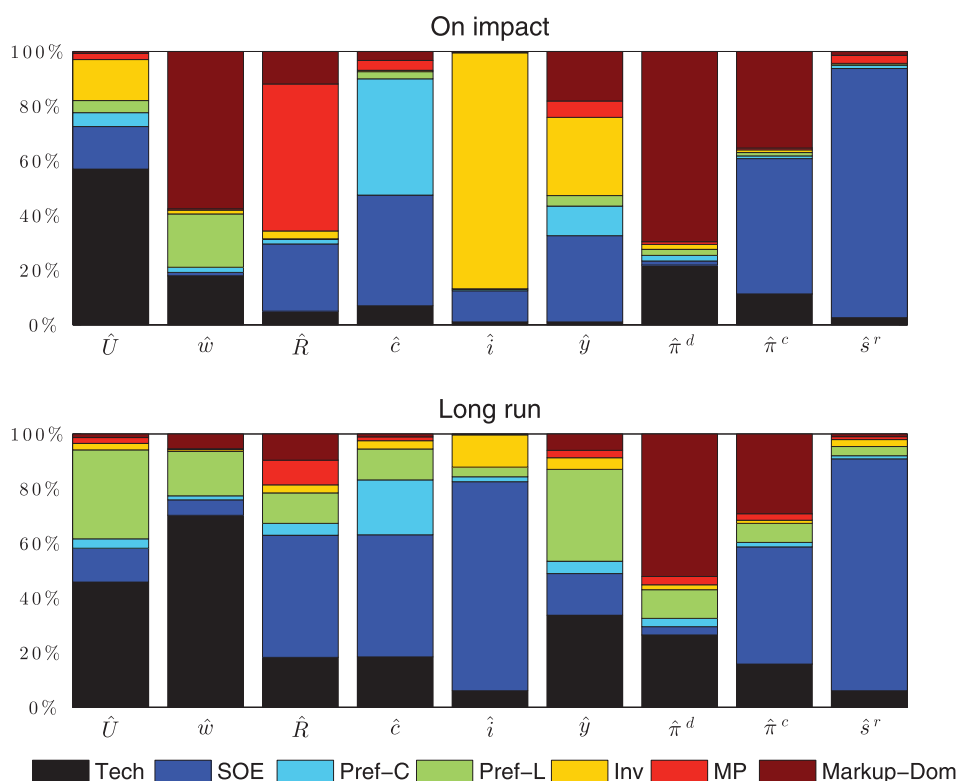


Fig. 5. Variance decomposition.

The variance of CPI inflation $\hat{\pi}^c$ is strongly dominated by open economy and technology shocks both in the short and long run. Domestic goods inflation ($\hat{\pi}^d$) variance is heavily affected by domestic markup shocks (almost 70% in the short run, 52% in the long).

The variance of (unexpected) monetary policy shocks only has an important impact on the short-term nominal interest rate ($\hat{R}$) variance. This indicates the Reserve Bank of Australia has communicated its intentions effectively, and has not caused significant surprises to the Australian economy, even in the short run.

If hiring costs are assumed away, there is a limited impact of technology shocks in both the short and long run. This in turn means that technology shocks have no role in driving the GDP deflator and the CPI inflation. Therefore, without hiring costs that bridge the temporary technology shocks to labour demand, the model attributes a minimal role for technology shocks in driving the real wages and inflation.²⁹

4.3. Conditional forecasts

We now examine the out-of-sample performance of our preferred model, focusing on how well it does during the global financial crisis. We estimate the model using data up to 2007Q3, then forecast the series to 2013Q2.³⁰ Since this crisis was external to Australia, our conditional forecasts are calculated by imposing the actual data paths for foreign interest rates, foreign inflation, foreign output growth and exchange rate in the forecast period. In addition, since Australia experienced a boom in the real value of its mining exports from the mid-2000s mainly due to an ever-growing China, we also impose the actual values of export growth on our conditional forecasts.

Fig. 6 presents the actual data series (red solid lines) and our preferred model's conditional forecasts (blue dashed lines) from 2007Q4 to 2013Q2 with 90% credible bands (thin solid lines). Despite the obvious difficulties that such a model faces in predicting well in such a major global crisis, the forecast values of all the key macroeconomic variables after 23 quarters are statistically close to their actual values apart from the interest rate (which was much lower than forecast, largely because relevant world interest rates were so low). The root mean square errors (shown in the sub-headers of each pane in Fig. 6) are remarkably low for the main domestic macroeconomic variables. The model fails to pick up the large fall in output

²⁹ Without hiring costs, technology shocks explain less than 1% (11%) of real wage variance in the short(long) run. The details of the variance decomposition are available from the authors upon request.

³⁰ We omit the estimation results due to space, they are very similar to those estimated using the full sample, and are available from the authors upon request.

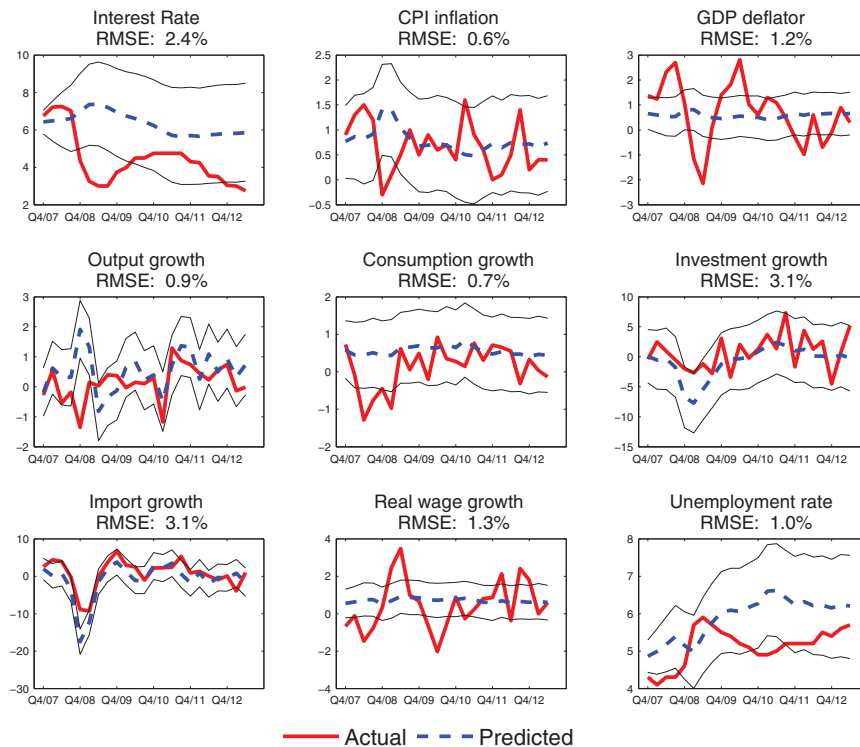


Fig. 6. Conditional forecasts.

and consumption growth through 2008. This is unsurprising since these are likely explained by the collapse in consumer confidence that actually occurred.³¹ For similar reasons, our model fails to predict the sharp decline in the cash rate in 2008. The immediate and large response of monetary policy and of fiscal policy (which we did not model), and the boom in mining exports helped the Australian economy to recover from this crisis in confidence. From then onwards, the model's conditional forecasts were remarkably good for all domestic macroeconomic variables.

Our model predicts too smooth a path for real wage growth and the GDP deflator. This is consistent with our findings from variance decomposition: fluctuations of these variables are mainly driven by domestic market shocks. On the contrary, external factors are important in determining unemployment. Our conditional forecasts indicate that without the fast response of monetary policy (as well as fiscal policy which we did not model), the unemployment rate may have been much higher than otherwise for the four years after the crisis. Aside from confidence effects, the model successfully predicts the relatively small impact that the global financial crisis had on the Australian economy.

5. Conclusions

In this paper, we estimate a medium-scale small open economy DSGE model with labor market frictions using quarterly Australian data from 1993 to 2013. We evaluate four alternatives of this model: with nominal or real wage rigidity and/or with and without hiring costs.

Real wage rigidity allows a direct role for hiring costs to determine real wages through Nash bargaining, but has no direct effect under nominal wage rigidity, which instead allows GDP deflator and CPI inflation to play larger roles in driving real wages. Our results confirm the existence of hiring costs in the Australian labor market, with the point estimates suggesting they account for 1.58% of GDP, with a 90% credibility interval ranging from 0.85 to 2.26. Using the log of the Bayes factor as a guide, the Australian data prefers nominal wage rigidity and the inclusion of hiring costs compared to the other paired alternatives. In other words, the model with both nominal wage rigidity and hiring costs best fits the data.

Variance decompositions for our preferred model indicate that labor supply shocks are important for real wages in the short run and output, real wages and unemployment in the long run. Technology shocks are equally important for unemployment in the short and long run, but plays a much bigger role for real wages in the long run than in the short run. Output variations are driven by aggregate demand shocks in the short run, and by aggregate supply shocks in the long run,

³¹ The Westpac-Melbourne Institute consumer sentiment index fell from 112.5 in December 2007 to 79 in July 2008, recovering only in mid-2009.

thus validating the New Keynesian description of our model. A key result is that movements in key labour market variables are minimally driven by external shocks.

The out-of-sample conditional forecast performance of the model shows that it was difficult to predict the immediate impact of the global financial crisis, which were probably due to confidence effects. However, once these immediate effects had dissolved (probably on account of the large response of monetary policy, the un-modeled fiscal stimulus and the mining export boom), our model performed remarkably well in predicting domestic macroeconomic variables four years after the crisis began. We conclude that our New Keynesian model remains valid for this economy despite the global crisis.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.jmacro.2016.02.006](https://doi.org/10.1016/j.jmacro.2016.02.006)

References

- Adolfson, M., Laseen, S., Linde, J., Villani, M., 2007. Bayesian estimation of an open economy DSGE model with incomplete pass-through. *J. Int. Econ.* 72 (2), 481–511.
- Albertini, J., Kamber, G., Kirker, M., 2012. Estimated Small Open Economy Model With Frictional Unemployment. Technical Report 2. Australian Bureau of Statistics, 2003. Employment services 2001–2002, cat. no. 8558.0, ABS, Canberra.
- Basu, S., Fernald, J.G., Kimball, M.S., 2006. Are technology improvements contractionary? *Am. Econ. Rev.* 96 (5), 1418–1448.
- Blanchard, O., Gali, J., 2007. Real wage rigidities and the new keynesian model. *J. Money Credit Bank.* 39, 35–65.
- Blanchard, O., Gali, J., 2010. Labor markets and monetary policy: a new Keynesian model with unemployment. *Am. Econ. J. Macroeconomics* 2 (2), 1–30.
- Buncic, D., Melecky, M., 2008. An estimated new-keynesian policy model for australia. *The Econ. Rec.* 84 (264), 1–16.
- Calvo, G.A., 1983. Staggered prices in a utility-maximizing framework. *J. Monetary Econ.* 12 (3), 383–398.
- Christiano, L.J., Trabandt, M., Walentin, K., 2011. Introducing financial frictions and unemployment into a small open economy model. *J. Econ. Dyn. Control* 35 (12), 1999–2041.
- Clarida, R., Gali, J., Gertler, M., 1999. The science of monetary policy: a New Keynesian perspective. *J. Econ. Lit.* 37 (4), 1661–1707.
- Clarida, R., Gali, J., Gertler, M., 2000. Monetary policy rules and macroeconomic stability: Evidence and some theory. *Q. J. Econ.* 115 (1), 147–180.
- Faccini, R., Millard, S., Zanetti, F., 2013. Wage rigidities in an estimated dynamic, stochastic, general equilibrium model of the UK labour market. *Manchester School* 81, 66–99.
- Francis, N., Ramey, V.A., 2005. Is the technology-driven real business cycle hypothesis dead? shocks and aggregate fluctuations revisited. *J. Monetary Econ.* 52 (8), 1379–1399.
- Gali, J., 1999. Technology, employment, and the business cycle: Do technology shocks explain aggregate fluctuations? *Am. Econ. Rev.* 89 (1), 249–271.
- Gali, J., Monacelli, T., 2005. Monetary policy and exchange rate volatility in a small open economy. *Rev. Econ. Stud.* 72 (3), 707–734.
- Gali, J., van Rens, T., 2010. The vanishing procyclicality of labor productivity. IZA Discussion Papers 5099. Institute for the Study of Labor (IZA).
- Gali, J., Smets, F., Wouters, R., 2011. Unemployment in an estimated new keynesian model. NBER Macroeconomics Annual 2011, Volume 26. National Bureau of Economic Research, Inc. NBER Chapters.
- Gertler, M., Sala, L., Trigari, A., 2008. An estimated monetary DSGE model with unemployment and staggered nominal wage bargaining. *J. Money Credit Bank.* 40 (8), 1713–1764.
- Goodfriend, M., 1991. Interest rates and the conduct of monetary policy. *Carnegie-Rochester Conf. Ser. Pub. Pol.* 34 (1), 7–30.
- Guerron-Quintana, P.A., 2010. What you match does matter: the effects of data on DSGE estimation. *J. Appl. Econometrics* 25 (5), 774–804.
- Hall, R.E., 2003. Wage determination and employment fluctuations. NBER Working Papers 9967. National Bureau of Economic Research, Inc.
- Hall, R.E., 2005. Employment fluctuations with equilibrium wage stickiness. *Am. Econ. Rev.* 95 (1), pp. 50–65.
- Jääskelä, J., Nimark, K., 2011. A medium-scale new Keynesian open economy model of Australia. *Econ. Rec.* 87 (276), 11–36.
- Kass, R.E., Raftery, A.E., 1995. Bayes factors. *J. Am. Stat. Assoc.* 90 (430), pp. 773–795.
- Levin, A., Wieland, V., Williams, J., 1999. Robustness of simple monetary policy rules under model uncertainty. In: Taylor, J.B. (Ed.), *Monetary Policy Rules*. University of Chicago Press, NBER Chapters, pp. 263–318.
- Nimark, K.P., 2009. A structural model of australia as a small open economy. *Australian Econ. Rev.* 42 (1), 24–41.
- Ponomareva, N., Sheen, J., 2010. Cyclical flows in australian labour markets. *Econ. Rec.* 86, 35–48.
- Riggi, M., 2010. Nominal and real wage rigidities in new keynesian models: a critical survey. *J. Econ. Surv.* 24, (3), 539–572.
- Riggi, M., Tancioni, M., 2010. Nominal vs real wage rigidities in new keynesian models with hiring costs: a bayesian evaluation. *J. Econ. Dyn. Control* 34 (7), 1305–1324.
- Sack, B., Wieland, V., 2000. Interest-rate smoothing and optimal monetary policy: a review of recent empirical evidence. *J. Econ. Bus.* 52 (1–2), 205–228.
- Schmitt-Grohe, S., Uribe, M., 2003. Closing small open economy models. *J. Int. Econ.* 61 (1), 163–185.
- Sheen, J., Wang, B.Z., 2014. An Estimated Small Open Economy Model with Labour Market Frictions. Dynare Working Papers 35. CEPREMAP.
- Shimer, R., 2005. The cyclical behavior of equilibrium unemployment and vacancies. *Am. Econ. Rev.* 95 (1), 25–49.
- Smets, F., Wouters, R., 2007. Shocks and frictions in US business cycles: a Bayesian DSGE approach. *Am. Econ. Rev.* 97 (3), 586–606.
- Woodford, M., 1999. Optimal monetary policy inertia. *Manchester School* 67, 1–35.